

Proving Threshold Regev PKE from Adaptive Hint-MLWE: Efficient, Non-interactive, and CCA Secure

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Abstract. Threshold public-key encryption (tPKE) has recently attracted renewed interest, largely due to NIST’s call for Multi-Party Threshold Cryptography. While classical tPKE has approached a high state of maturity, its post-quantum counterpart has not. Indeed, thresholdizing the celebrated lattice-based Regev PKE, which forms the basis of ML-KEM, remains unsatisfactory. Interestingly, *how* to thresholdize Regev PKE has not fundamentally changed in over a decade — the only thing that has gradually progressed is its security analysis. To this day, it remains open whether threshold Regev can be proven secure while simultaneously satisfying a polynomial modulus, non-interactive decryption, and CCA-compatibility, each of which is essential for practical deployment.

We answer this affirmatively, providing the first proof that threshold Regev is secure under the MLWE assumption while satisfying all three requirements. In fact, we prove that it satisfies a very strong form of simulation-based security — even stronger than what was known under a super-polynomial modulus — allowing the adversary to obtain partial decryptions even of the challenge ciphertext. At the technical heart of our result is the *adaptive* hint-MLWE (AHMLWE) problem, an adaptive variant of hint-MLWE where the adversary obtains hints on the MLWE secret with *adaptively* chosen coefficients. We show that AHMLWE reduces tightly to standard MLWE, which may be of independent interest.

1 Introduction

A t -out-of- n threshold public-key encryption (tPKE) strengthens the security of traditional PKE in sensitive deployments by distributing the secret key across n parties so that any t of them can jointly decrypt, while no single participant can act alone. While tPKE has always been a topic of interest, it has seen renewed interest in recent years, notably due to NIST announcing a call for threshold cryptosystems [BP26]. Classical applications include preserving voter privacy in e-voting [CGS97, ABGS23], while more modern applications include mempool privacy for blockchains and decentralized finance [CGPP24, CGPW25].

In the classical setting, tPKE has approached a high state of maturity, with efficient threshold variants of RSA and ElGamal known for decades. In contrast, constructing an efficient and secure tPKE from post-quantum assumptions has proven to be challenging. While the study of lattice-based tPKE was initiated over a decade ago by Bendlin and Damgård [BD10], the community is still making progress, proposing numerous constructions to this date, e.g., [BGG⁺18, KLO⁺19, DLN⁺21, CCMS21, BS23, MS24, CLW25, LP25, BdLP26].

In this work, as with many prior works, our target is thresholdizing the Regev PKE [Reg05], or more precisely, its optimized variant [LPR10, LP11], which forms the basis of ML-KEM [Nat24].⁴ Our goal is to satisfy all of the following requirements *simultaneously*, each of which is essential for practical deployment:

1. *Polynomial-sized modulus*: the scheme can be instantiated with concretely efficient parameters;
2. *(Asynchronous) non-interactive decryption*: each party outputs a single partial decryption without knowing the set of decryptors;
3. *CCA-compatible*: the scheme can be generically upgraded to CCA security.

Below, let us take a closer look at why these seemingly modest requirements have been elusive to achieve at the same time.

Blueprint for CPA-secure threshold Regev. While there are subtle differences, all threshold Regev PKEs share the same blueprint. The public key consists of a matrix A and $\vec{b} = -A\vec{s} + \vec{e} \bmod q$ for some short noise vector $\vec{e}$, and a ciphertext of a message m takes the form $(c_0, \vec{c}_1) = (\vec{r}_2^\top \vec{b} + r_0 + \text{Encode}(m), A^\top \vec{r}_2 + \vec{r}_1) \bmod q$ for some short noise $r = (r_0, \vec{r}_1, \vec{r}_2)$ and encoding algorithm Encode . The secret $\vec{s}$ is linearly secret-shared among the parties into shares $(\vec{s}_i)_{i \in [n]}$, where for any valid set of decryptors $\mathcal{S}$, there exist constants $\lambda_i^\mathcal{S}$ such that $\vec{s} = \sum_{i \in \mathcal{S}} \lambda_i^\mathcal{S} \vec{s}_i$.

To perform non-interactive decryption, each party outputs its partial decryption $\vec{c}_1^\top \vec{s}_i + f_i \bmod q$ for “some” noise f_i to be discussed later. Here, note that without the noise f_i , one can trivially recover $\vec{s}_i$ using Gaussian elimination given sufficiently many partial decryptions. Once partial decryptions from a valid set of decryptors $\mathcal{S}$ are collected, we can compute $c_0 + \sum_{i \in \mathcal{S}} \lambda_i^\mathcal{S} \cdot \vec{c}_1^\top \vec{s}_i \approx \text{Encode}(m) + \sum_{i \in \mathcal{S}} \lambda_i^\mathcal{S} f_i \bmod q$. Therefore, as long as the secret sharing scheme supports small reconstruction coefficients $\lambda_i^\mathcal{S}$, we can recover m .⁵

How to prove CPA security. The main difference between all prior works on threshold Regev boils down to how this blueprint is proven secure. While there is a hierarchy of CPA security notions for tPKE [BKW26], at a minimum, the

⁴ Throughout the introduction, we do not differentiate between these variants for simplicity, as they are interchangeable in the context of tPKE.

⁵ It is well-known that this restriction on $\lambda_i^\mathcal{S}$ can be removed in the “synchronous” setting, where the set of decryptors is known at the time of partial decryption (e.g., [dVPS26]).

scheme must tolerate partial decryption queries on honestly generated, known-message ciphertexts ct_m . That is, the adversary is given access to an encryption oracle EncO , which on input m outputs an honestly generated ciphertext ct_m , and it may then obtain partial decryptions of ct_m . This implicitly models the requirement that the secret shares $\tilde{s}_i$ do not leak.

The initial proof by Bendlin and Damgård [BD10] showed that threshold Regev is secure in the n -out-of- n setting if f_i is super-polynomially larger than the encryption noise r , leading to a scheme with a super-polynomial modulus q . Later, [BGG⁺18, DLN⁺21] showed, among others, that threshold Regev is secure even in the t -out-of- n setting, still assuming a super-polynomial q , to the security parameter λ and the number of queries Q .

The first to prove threshold Regev secure under a *polynomial* modulus was Boudgoust and Scholl [BS23]. However, as they relied on the Rényi divergence⁶, which does not directly apply to decision-based games, they were only able to prove OW-CPA security — to achieve IND-CPA security, they had to modify threshold Regev. Later, Micciancio and Suhl [MS24] also proved (a slight variant of) threshold Regev secure under a polynomial modulus, but restricted to the n -out-of- n and unstructured LWE setting; that is, their result does not extend to the structured module setting without assuming a non-standard *known-norm* MLWE assumption. The state-of-the-art is the recent work by Cini, Lai, and Woo [CLW25], who proved that threshold Regev (which they call Pilvi) is secure under a polynomial modulus by introducing an a priori bound Q_{enc} on the number of EncO queries, with their modulus scaling as $q \approx \sqrt{Q_{\text{enc}}}$. Unlike a bound on the number of partial decryptions, however, this is an artificial bound that is unobservable in the real-world. Counterintuitively, even if the adversary does *not* obtain any partial decryption of ct_m , the proof still requires the modulus of threshold Regev to scale with Q_{enc} .

Towards CCA security. Beyond CPA security, achieving CCA security is essential for the practical applicability of tPKE. For standard PKEs, we typically rely on the Fujisaki-Okamoto (FO) transform [FO99] to convert a CPA-secure PKE into a CCA-secure one. However, as the FO transform relies on a re-encryption technique that checks the validity of the ciphertext using the underlying plaintext, it is notoriously difficult to perform in a threshold manner. Indeed, existing works thresholdizing the FO transform rely on generic MPC techniques [KLO⁺19, CCMS21], leading to a highly inefficient interactive decryption protocol.

Another popular approach is the encrypt-and-prove methodology, where the encryptor attaches a proof π computed via a non-interactive zero-knowledge proof (NIZK), demonstrating knowledge of the ciphertext randomness. Unlike the FO transform, the parties can simply perform threshold decryption of the CPA-secure tPKE after *publicly* checking the proof π , making it “seemingly”

⁶ They assume an a priori upper bound Q on the number of partial decryptions the adversary obtains, a convention that has since remained widely used in tPKE. Limiting the number of oracle queries in security proofs is a common convention that predates this work (e.g., [ASY21] for threshold signatures).

compatible with CCA security. Indeed, many prior works (e.g., [BS23, MS24, CLW25]) suggested this as a viable path. However, none of them provided a formal proof, nor a concrete instantiation of a CCA-secure tPKE.

Recently, [BKW26] exposed the pitfall of this approach: they provided a hierarchy of CPA security notions and showed that, depending on the level of CPA security satisfied by the underlying tPKE, the above NIZK transform does *not* result in a CCA-secure scheme. At a high level, this is because the reduction R (i.e., the adversary against the CPA-secure tPKE) must simulate the partial decryptions of the honest parties to the adversary in the CCA security game. Observe that this is unique to the threshold setting: in the non-threshold setting, if R can extract the message from the proof π attached to an adversarially submitted ct_A , R can easily simulate the decryption oracle by returning m .

Let us go back to the minimal CPA security notion discussed above. In that game, the CPA adversary has access to an EncO oracle outputting *honestly generated* ct_m , and a PartDecO oracle that on input ct_m outputs the partial decryptions of the honest parties. It is clear that this is useless to R , as it offers no interface to obtain partial decryptions of adversarial but well-formed ciphertexts ct_A . Indeed, what R needs is an EncO oracle that allows it to specify the (potentially malicious) randomness used to generate ct_m . [BKW26] coined such security *semi-malicious* CPA security. They also defined a slightly weaker notion where EncO returns the (honestly generated) randomness used to construct ct_m , along with a generic transform relying instead on a non-interactive *proof of randomness* (NIPoR) — as a NIPoR is a stronger primitive than a NIZK, this results in a less efficient CCA transform.

Summary of the situation. It turns out that the only security proofs of threshold Regev covering semi-malicious CPA security are the initial ones [BD10, BGG⁺18], both relying on a super-polynomial modulus.⁷ Notably, all subsequent works focusing on a polynomial modulus [BS23, MS24, CLW25] are *not* compatible with the NIZK transform, effectively *trading security with efficiency*. While [CLW25] claims that the NIPoR transform of [BKW26] can be used on Pilvi to achieve CCA security, this is not the case upon closer inspection, since they achieve a strictly weaker notion of CPA security than what the transform requires.⁸

Therefore, while there has been significant progress in our understanding of threshold Regev, we still do not know the answer to the following fundamental question, which brings us back to the main goal of this work:

*Can the Regev PKE be safely and efficiently thresholdized,
simultaneously satisfying requirements (1) to (3)?*

⁷ While these works do not claim semi-malicious CPA security, it is clear that their proofs achieve it. We ignore [DLN⁺21] as it is a direct construction of a CCA-secure scheme, even though it follows the general NIZK transform.

⁸ We informed the authors of this issue, and they have confirmed that their weak notion of CPA security is insufficient for achieving CCA security with polynomial modulus. Accordingly, their concrete parameters in Section 7 is limited to a CPA secure scheme.

1.1 Our Contribution

In this work, we answer the above question affirmatively. We provide the first proof that threshold Regev is secure under the MLWE assumption while satisfying all three requirements: a polynomial modulus (Item 1), non-interactive decryption (Item 2), and CCA-compatibility (Item 3). In fact, we prove that threshold Regev satisfies the strongest *simulation-based* CPA security in the hierarchy of [BKW26] — even stronger than what was proven by [BD10, BGG⁺18] under a super-polynomial modulus — allowing the adversary to obtain partial decryptions even of the *challenge* ciphertext. A comparison with prior CPA-secure schemes is provided in Table 1.

Work	Polynomial modulus	PartDecO on		rnd. Model		CCA compatible
		L_{Chal}	L_{Enc}			
[BD10] [†]	✗	✗	✓	mal.	SIM	✓
[BGG ⁺ 18]	✗	✗	✓	mal.	SIM	✓
[BS23]	✓	✗	✓	⊥	IND	✗
[MS24] ^{†,‡}	✓	✗	✓	⊥	SIM	✗
[CLW25]	✓	✓ ⁽⁻⁾	✓ ⁽⁻⁾	hon.	SIM	✗
Ours	✓	✓	✓	mal.	SIM	✓

Table 1. Comparison of CPA-secure (Regev-style) non-interactive tPKEs. Oracles L_{Chal} and L_{Enc} provide partial decryptions of challenge ciphertexts ct^* and known-message ciphertexts ct_m , respectively. Column “rnd.” indicates ct_m randomness: hidden ($\perp$), honest (hon. and given to $\mathcal{A}$), or malicious (mal. and chosen by $\mathcal{A}$). IND and SIM denote indistinguishability and simulation-based security. A scheme is CCA-compatible if it can be transformed into a CCA-secure one via a generic NIZK transform. [†]: Supports only the n -out-of- n setting ($t = n$). [‡]: [MS24] requires non-standard MLWE assumption. ⁽⁻⁾: [CLW25] bounds the number of ct_m created, preventing the CCA transform from [BKW26]. They also restrict the number of partial decryptions of ct^* to $t - 1$ parties.

At the technical heart of our result is the *adaptive* hint-MLWE (AHMLWE) problem, an adaptive variant of the hint-MLWE problem [KLSS23] that allows the adversary to obtain hints on the MLWE secret with *adaptively* chosen coefficients. We show that AHMLWE reduces tightly to the standard MLWE problem using the noise-rerandomization technique of Katsumata and Yamada [KY16]. Our security proof then consists of rewriting the partial decryptions of the honest parties and carefully layering the hybrid games, invoking several variants of AHMLWE in order.

We further provide the first concrete instantiation of a CCA-secure threshold Regev PKE by instantiating the NIZK with the Lantern proof system [LNP22]. For concrete efficiency, we rely on Shamir secret sharing over the monomi-

als [BFM⁺25, CLW25, KLL⁺26], supporting up to $n = 32$ parties.⁹ Table 2 reports concrete ciphertext sizes for both our CPA- and CCA-secure variants across a range of thresholds t (see Sec. 6, Table 4 for more details).

mode \ t	2	16	31
CPA	19.10	43.97	19.41
CCA	104.91	177.48	105.49

Table 2. Ciphertext size in kilobytes (KB) when the number of parties $n = 32$, the number of decryption queries $Q = 2^{60}$.

1.2 Related Work

Threshold fully homomorphic encryption. A closely related line of work constructs threshold *fully homomorphic* encryption (tFHE). Boneh et al. [BGG⁺18] proposed the first tFHE under super-polynomial noise. Unlike tPKE, tFHE does not need to consider CCA security. Since then, many papers have followed, e.g., [BS23, PS24, OT25, ZZLP25a, OT26]. The recent lines of work [PS24, OT25, OT26] have focused on a modification of [BGG⁺18] where they optimize the size of the partial decryption. While ciphertext ct still lives over a super-polynomial modulus, they include a “sanitization” step so that the partial decryption lives over a polynomial modulus. Some works additionally introduce multi-party computation (MPC) for the decryption to achieve a polynomially large modulus, however, they are either confined to a 3-party setting [KLO⁺19] or require a power-of-two modulus to work [ZZLP25b]. Most importantly, they do not support a constant-round decryption.

More CCA-secure tPKEs. Devevey et al. [DLN⁺21] construct the first *adaptively* and CCA secure lattice-based tPKE in the *standard* model (i.e., no random oracles). They rely on lossy arguments, requiring a super-polynomial sized modulus, and mainly serving as a feasibility result. Micciancio and Suhl [MS24] build the only threshold Regev that achieves a polynomial modulus without any restrictions on the number of partial decryptions. Their proof explicitly relies on the weaker CPA model, where the encryption randomness is hidden from the adversary — combining their proof strategy with ours will be an interesting research direction. Recently, Lapiha and Prest [LP25] and Boudgoust et al. [BdLP26] departed from the threshold Regev blueprint and focused directly on constructing a lattice-based CCA-secure tPKE with polynomial modulus. Their construction is based on building a CPA-secure threshold identity-based encryption (tIBE)

⁹ Asymptotically, our proof supports an unbounded polynomial number of parties by relying on $\{0, 1\}$ -linear secret sharing schemes (e.g., [BGG⁺18, BS23]). However, since these are concretely inefficient, we did not choose them for our instantiation.

and converting it into a CCA-secure tPKE via a threshold variant of the BCHK transform [CHK04, BCHK07]. They achieve a small ciphertext size of about 30 kB for $t = 32$ and up to $Q = 2^{46}$ decryption queries, however at the cost of requiring an *interactive* decryption protocol. Moreover, it is unclear if their construction can achieve a strong form of simulation security, supporting partial challenge ciphertext decryptions.

1.3 Technical Overview

We now describe our two main technical contributions: the adaptive hint-MLWE problem and its reduction to MLWE, and how they drive the security proof of our tPKE scheme, including the simulation of partial decryptions of *challenge* ciphertexts. Throughout, $\mathcal{R} := \mathbb{Z}[X]/(X^N + 1)$ denotes the ring of integers of the $2N$ -th cyclotomic field for a power-of-two integer N , and $\mathcal{R}_q := \mathcal{R}/q\mathcal{R}$ for an integer modulus q ; vectors and matrices over $\mathcal{R}$ are written in bold.

Adaptive hint-MLWE. The MLWE problem asks to distinguish $(A, [I_k \mid A] \cdot \vec{x})$ from a uniformly random pair, for $A \xleftarrow{\$} \mathcal{R}_q^{k \times \ell}$ and $\vec{x} \leftarrow \chi^{k+\ell}$. Hint-MLWE [KLSS23] strengthens the adversary by additionally revealing linear hints $\vec{v}_i^\top \vec{x} + y_i$ on the secret $\vec{x}$, where the coefficient vectors $\vec{v}_i$ are fixed *before* the game begins and y_i is drawn from an independent noise distribution χ' .

In our setting, each partial decryption is, after a short rewriting, exactly such a hint. Substituting the public-key and ciphertext relations $\vec{b} = -A\vec{s} + \vec{e}$ and $\vec{c}_1 = A^\top \vec{r}_2 + \vec{r}_1$ reveals a linear function of the *short* secret $(\vec{s}, \vec{e})$, with the ciphertext's own encryption randomness $r = (r_0, \vec{r}_1, \vec{r}_2)$ as its coefficient and flooding noise f_i . This fits selective hint-MLWE only when the encryption randomness is fixed in advance — but a semi-malicious adversary chooses it adaptively, so the relevant coefficient is only known when the query is made. We therefore define *adaptive hint-MLWE* (AHMLWE, Def. 9), where the adversary may query an oracle $\mathcal{O}_{\text{hint}}$ on adaptively chosen inputs $\vec{v} \in \mathcal{V}$ and receive $\vec{v}^\top \vec{x} + y$ for fresh $y \leftarrow \chi'$.

Hardness is not a direct corollary of the selective case, since a reduction to hint-MLWE must commit to its hints before seeing the eventual queries. We rely on the noise-rerandomization technique of Katsumata and Yamada [KY16] (Lem. 7): from one selective hint, one can locally compute $M\vec{x} + \vec{y}'$ for any bounded matrix M , with $\vec{y}'$ statistically close to a fresh Gaussian. Our reduction (Thm. 1) asks for one selective hint per secret coordinate and answers each adaptive query by rerandomizing the appropriate linear combination of these hints, keeping the loss relative to the underlying hint-MLWE instance tight.

The threshold Regev blueprint, revisited. Our scheme follows the threshold Regev blueprint (Fig. 3). The public key is $(\vec{b}, A)$ for $A \leftarrow \mathcal{U}(\mathcal{R}_q^{k \times k})$ and $\vec{b} = -A\vec{s} + \vec{e}$, with $\vec{s}, \vec{e} \leftarrow D_{\sigma_s}^k$; a ciphertext of m is $(c_0, \vec{c}_1)$ for $c_0 = \vec{r}_2^\top \vec{b} + r_0 + \gamma^{-1} \lfloor q/p \rfloor m$ and $\vec{c}_1 = A^\top \vec{r}_2 + \vec{r}_1$, where $(r_0, \vec{r}_1, \vec{r}_2) \leftarrow D_{\sigma_r}$; and the secret $\vec{s}$ is Shamir-shared at monomial evaluation points into shares $\vec{s}_1, \dots, \vec{s}_n$, so that $\vec{s} = \sum_{i \in \mathcal{I}} \lambda_i^T \vec{s}_i$ for any size- t set $\mathcal{I}$. To support asynchronous decryption — where

the set of decryptors is not known when partial decryptions are produced — each party simply releases the unweighted $d_i = \vec{c}_1^\top \vec{s}_i + f_i$; the Lagrange weights and a clearing factor γ — ensuring $\gamma \lambda_i^{\mathcal{I}}$ is small — only enter at the very end, in FinDec.

Semi-malicious security via adaptive hint-MLWE. The proof proceeds in three stages. First, we show that partial decryptions leak nothing about the secret key itself, even though they may still leak the message — which is harmless at this point, since no ciphertext has yet been declared a challenge. Second, we show that challenge ciphertexts can themselves be replaced by uniformly random values, so that they stop leaking their message. Third, and this is genuinely specific to the threshold setting, we must still answer partial-decryption queries on these now-message-independent challenge ciphertexts: one more application of AHMLWE turns certain virtual shares into one-time masks, after which a purely combinatorial argument about Shamir sharing, taken up in the next paragraph, ensures that the adversary’s view of partial decryptions of challenge ciphertexts can be simulated without knowing the message when fewer than t queries are made.

For the first stage, fix a maximal corrupted set $\mathcal{C}' \supseteq \mathcal{C}$ of size $t - 1$. By Lagrange interpolation against $\mathcal{C}'$, the share $\vec{s}_j$ of an honest party $j \notin \mathcal{C}'$ satisfies $\vec{s}_j = \mu_j \cdot (\vec{s} - \sum_{k \in \mathcal{C}'} \lambda_k^j \vec{s}_k)$ for $\mu_j := (\lambda_j^{\mathcal{C}' \cup \{j\}})^{-1}$, so its partial decryption $d_j = \vec{c}_1^\top \vec{s}_j + f_j$ reduces, via $\vec{b} = -A\vec{s} + \vec{e}$ and $\vec{c}_1 = A^\top \vec{r}_2 + \vec{r}_1$, to a known part plus the single term $\mu_j(\vec{r}_1^\top \vec{s} + \vec{r}_2^\top \vec{e}) + f_j$ — exactly a hint on $(\vec{s}, \vec{e})$ with coefficient $\mu_j \cdot (\vec{r}_1, \vec{r}_2)$. Since a semi-malicious query lets $\mathcal{A}$ pick $\vec{r}$ itself, interleaved with other queries, this coefficient is only fixed once the ciphertext is produced — precisely the adaptivity AHMLWE is built to tolerate, and where selective hint-MLWE would not suffice. Once AHMLWE replaces $\vec{b}$ by uniform, every partial decryption carries no information about the secret, regardless of which ciphertexts it decrypts, challenge or not.

For the second stage, a symmetric argument handles each challenge ciphertext itself: reading it as an MLWE sample with secret $\vec{r}_2$ and small error $(r_0, \vec{r}_1)$, the same partial-decryption term now leaks its randomness as a hint with coefficient $\vec{v} = \mu_j \cdot (1, \vec{s}, \vec{e})$ — the honestly sampled $\vec{s}, \vec{e}$ playing the role that the encryption randomness played above. After applying AHMLWE, the challenge ciphertexts can be replaced by uniform, so that they carry no information about their message.

For the third stage, what remains is to actually answer partial-decryption queries on these challenge ciphertexts independently of the message — and a final application of the same idea (deferred to the proof of Thm. 3) sets this up. It targets the virtual shares of $\mathcal{C}' \setminus \mathcal{C}$: parties that are never corrupted. The key observation is that every dependency on such a share $\vec{s}_k$ — in the partial decryption of *any* ciphertext, challenge or not — appears through exactly two quantities: an MLWE sample $\vec{b}_k = A\vec{s}_k + \vec{e}_k$, playing the role of a “virtual public key” for party k , and, for each challenge ciphertext ct , a value $d_{\text{ct},k} = \vec{c}_1^\top \vec{s}_k + f_{\text{ct},k}$, playing the role of party k ’s own partial decryption of ct . Replacing both by uniform via AHMLWE (after pulling out a factor of γ to keep the relevant

coefficients small) turns each $d_{\text{ct},k}$, for $k \in \mathcal{C}' \setminus \mathcal{C}$, into what is effectively a fresh one-time mask on the partial decryptions of honest parties for ct — exactly the missing ingredient for simulating partial decryptions of challenge ciphertexts, which we turn to next.

Simulating challenge partial decryptions. This last step needs no further hardness assumption: it is a purely information-theoretic consequence of Shamir sharing. Fix a challenge ciphertext ct . After the rerandomizations above, the partial decryption (minus flooding noise $f_{\text{ct},i}$) of *every* party $i \in [n]$ — corrupted or not — is exactly the evaluation $Q_{\text{ct}}(\alpha_i)$ of a degree- $<t$ polynomial Q_{ct} over $\mathcal{R}_q$ whose constant term $Q_{\text{ct}}(0)$ encodes the message m . Q_{ct} is pinned at two kinds of points: its value at 0, which depends on m , and its values at $\mathcal{C}$, which the corrupted parties can compute themselves from their own shares. The remaining $t - 1 - |\mathcal{C}|$ points are exactly where the one-time masks $d_{\text{ct},k}$, for $k \in \mathcal{C}' \setminus \mathcal{C}$, introduced above land, and are therefore uniformly distributed and independent of everything else. Shamir’s threshold guarantee ensures that this uniformity would hold no matter which $t - 1 - |\mathcal{C}|$ points other than $\mathcal{C}$ and 0 we chose to query. So the simulator never needs to know Q_{ct} : it answers the first $t - 1 - |\mathcal{C}|$ queries of $\text{PartDecO}(\text{ct}, \cdot)$ on distinct honest parties with fresh uniform values. Only once the adversary’s queries plus $\mathcal{C}$ reach the threshold t does Q_{ct} become pinned down; at that point, the simulator requests the real m (as SemiMalSimCPA permits, Fig. 1) and interpolates the necessary values. In short, simulating partial decryptions of a challenge ciphertext amounts to lazily sampling one Shamir-shared polynomial per ciphertext.

Achieving CCA security. We also propose a CCA-secure upgrade of our tPKE scheme, by relying on the BKW transform [BKW26], which can be seen as a threshold variant of the Naor-Yung paradigm [NY90]. This construction allows us to transform our semi-malicious CPA-secure tPKE scheme into a CCA-secure one by leveraging a simulation extractable (SIMEXT) NIZK system. This transform appends a NIZK proof of well-formedness to the ciphertext, and the partial decryption algorithm verifies the proof before running the baseline partial decryption.

Given that the statement to be proven consists only of norm bounds and linear relations, the Lantern proof system [LNP22] is a natural choice to instantiate this NIZK. However, Lantern is only proven to be weakly simulation-sound [BBP23], which is not sufficient for our purpose. We upgrade it to a SIMEXT NIZK system by relying on ML-DSA [DKL⁺18] as a strongly unforgeable signature scheme, and a single-key Regev PKE, following the encryption-to-the-sky approach.

We provide concrete parameters and corresponding ciphertext sizes using Lantern via the LaZer library [LSS24].

Round optimality. Our CCA-secure scheme is a direct augmentation of the Regev blueprint: decryption remains non-interactive and asynchronous, with each party independently outputting a single partial decryption. By contrast, the only other practical CCA-secure constructions, based on the BCHK+ trans-

form [LP25, BdLP26], require a synchronous channel and three to five communication rounds.

2 Preliminaries

2.1 Notations

$|\cdot|$ denotes the cardinality of a set. For an integer k , $[k]$ denotes the index set $[k] = \{1, \dots, k\}$. Let N be a power of two, and let q be an integer. We denote the ring of integers of the $2N$ -th cyclotomic field by $\mathcal{R} = \mathbb{Z}[X]/(X^N + 1)$. For some (finite) sets $\mathcal{T}$ and $\mathcal{S}$, if $|\mathcal{T}| = t$ and $\mathcal{T} \subseteq \mathcal{S}$, we write $\mathcal{T} \subseteq_t \mathcal{S}$.

For a vector $\mathbf{v} = (v_i)_{0 \leq i < m} \in \mathbb{Z}^m$, the L^p -norm and L^∞ -norm are defined as $\|\mathbf{v}\|_p = \left(\sum_{i=0}^{m-1} |v_i|^p\right)^{1/p}$ and $\|\mathbf{v}\|_\infty = \max_{0 \leq i < m} |v_i|$, respectively. For a matrix $\mathbf{A} \in \mathbb{Z}^{m \times n}$, the matrix two-norm of $\mathbf{A}$ is defined as $\|\mathbf{A}\|_2 = \max_{\|\mathbf{v}\|_2=1} \|\mathbf{A}\mathbf{v}\|_2$. If a matrix $\mathbf{A} \in \mathbb{R}^{n \times n}$ is positive semidefinite, meaning that $\mathbf{x}^\top \mathbf{A} \mathbf{x} \geq 0$ for any $\mathbf{x} \in \mathbb{R}^n$, we write $\mathbf{A} \succeq 0$. Similarly, for $v = v_0 + v_1X + \dots + v_{N-1}X^{N-1} \in \mathcal{R}$, we define the L^p -norm and L^∞ -norm as $\|v\|_p = \left(\sum_{i=0}^{N-1} |v_i|^p\right)^{1/p}$ and $\|v\|_\infty = \max_{0 \leq i < N} |v_i|$. We extend the L^2 -norm to tuples $\vec{v} = (v_i)_{0 \leq i < \ell} \in \mathcal{R}^\ell$ via their coefficient embedding into $\mathbb{Z}^{N\ell}$, i.e., $\|\vec{v}\|_2 := \left(\sum_{i=0}^{\ell-1} \|v_i\|_2^2\right)^{1/2}$.

We denote the uniform distribution over a finite set $\mathcal{S}$ as $\mathcal{U}(\mathcal{S})$. For a real $B > 0$, we write $\mathcal{V}_B$ to denote the set of all tuples $\vec{v} = (v_i)_i \in \mathcal{R}^*$ (of any length, clear from context) such that $\sum_i \|v_i\|_1^2 \leq B$. For a distribution $\mathcal{X}$, we write $x \leftarrow \mathcal{X}$ to denote sampling x from $\mathcal{X}$.

We also recall the definition of statistical distance, and a useful lemma.

Definition 1 (Statistical Distance). For any two distributions $\mathcal{X}_1$ and $\mathcal{X}_2$ with the same support Ω , the statistical distance is defined as $\Delta_{SD}(\mathcal{X}_1, \mathcal{X}_2) = \frac{1}{2} \sum_{x \in \Omega} |\mathcal{X}_1(x) - \mathcal{X}_2(x)|$.

Lemma 1 (Cauchy-Schwarz bound for ring inner products). For any $\vec{u}, \vec{v} \in \mathcal{R}^\ell$, $\|\vec{u}^\top \vec{v}\|_\infty \leq \|\vec{u}\|_2 \cdot \|\vec{v}\|_2$.

Proof. For each coefficient index $0 \leq c < N$, reduction modulo $X^N + 1$ makes the c -th coefficient of $X^c \cdot w$ a signed permutation $\tau_c(w)$ of w 's coefficients, so the c -th coefficient of $\vec{u}^\top \vec{v}$ equals $\langle \vec{u}, (\tau_c(v_i))_i \rangle$ (viewing both as vectors in $\mathbb{Z}^{N\ell}$), with $\|(\tau_c(v_i))_i\|_2 = \|\vec{v}\|_2$. Cauchy-Schwarz then bounds this by $\|\vec{u}\|_2 \|\vec{v}\|_2$ for every c . $\square$

2.2 Gaussian Distribution

For the covariance matrix $\Sigma \succeq 0$ and $\mathbf{v}, \mathbf{c} \in \mathbb{Z}^n$, the Gaussian mass function $\rho_{\mu, \sqrt{\Sigma}}(\mathbf{v}) : \mathbb{Z}^n \rightarrow (0, 1]$ is defined as follows:

$$\rho_{\mathbf{c}, \sqrt{\Sigma}}(\mathbf{v}) = \exp(-(\mathbf{v} - \mathbf{c})^\top \Sigma^{-1} (\mathbf{v} - \mathbf{c}) / 2).$$

The discrete Gaussian distribution restricts ρ to a full-rank lattice $\Lambda \subset \mathbb{R}^n$ and renormalizes: with center $\mathbf{c} \in \mathbb{R}^n$ and covariance Σ , its mass function is

$$D_{\Lambda, \mathbf{c}, \sqrt{\Sigma}}(\mathbf{v}) = \frac{\rho_{\mathbf{c}, \sqrt{\Sigma}}(\mathbf{v})}{\sum_{\mathbf{x} \in \Lambda} \rho_{\mathbf{c}, \sqrt{\Sigma}}(\mathbf{x})}.$$

In the subscripts of both ρ and D , we omit $\mathbf{c}$ when $\mathbf{c} = \mathbf{0}$, write σ for $\sqrt{\Sigma}$ when $\Sigma = \sigma^2 \mathbf{I}_n$, and omit Λ when clear from context.

We also recall useful definitions and lemmas.

Definition 2 (Smoothing parameter [MR04]). *For an n -dimensional lattice Λ and positive real $\varepsilon > 0$, the smoothing parameter $\eta_\varepsilon(\Lambda)$ is the smallest s such that $\rho_{1/(s\sqrt{2\pi})}(\Lambda^* \setminus \{\vec{0}\}) \leq \varepsilon$.*

Lemma 2 (Infinity-norm tail bound, adapted from [GPV08, Lemma 4.2]). *For $x \leftarrow D_{\mathbb{Z}, \sigma}$ and $t > 0$, $\Pr[|x| \geq t\sigma] \leq 2e^{-t^2/2}$. Consequently, for $s = (s_i)_{0 \leq i < \ell} \leftarrow D_{\mathcal{R}, \sigma}^\ell$ and $\varepsilon \in (0, 1)$, taking a union bound over the $N\ell$ coefficients of s , $\Pr[\max_{0 \leq i < \ell} \|s_i\|_\infty \geq \rho(\varepsilon, \ell) \cdot \sigma] \leq \varepsilon$, where $\rho(\varepsilon, \ell) := \sqrt{2 \ln(2N\ell/\varepsilon)}$.*

Lemma 3 (Two-norm tail bound, [DKM⁺24, Lemma 3.4]). *Let $s \leftarrow D_{\mathcal{R}, \sigma}^\ell$. Then, $\Pr[\|s\|_2 \geq e^{1/4} \sigma \sqrt{N\ell}] \leq 2^{-N\ell/10}$.*

Corollary 1. *Let $s = (s_i)_{0 \leq i < \ell} \leftarrow D_{\mathcal{R}, \sigma}^\ell$. Then, $\Pr\left[\sum_{i=0}^{\ell-1} \|s_i\|_1^2 \geq e^{1/2} N^2 \ell \sigma^2\right] \leq 2^{-N\ell/10}$, since Cauchy-Schwarz gives $\|s_i\|_1 \leq \sqrt{N} \|s_i\|_2$ for each i , so $\sum_i \|s_i\|_1^2 \leq N \|s\|_2^2$, and the claim follows by squaring Lem. 3's bound on $\|s\|_2$.*

For $u \in \mathcal{R}$, recall that one-norm submultiplicativity gives $\|u \cdot s_i\|_1 \leq \|u\|_1 \|s_i\|_1$ for every i ; combined with Corollary 1, this lets us bound $\sum_i \|u \cdot s_i\|_1^2 \leq \|u\|_1^2 \sum_i \|s_i\|_1^2$ whenever u is independent of s .

2.3 Module Learning With Errors

We describe the Module Learning With Errors (MLWE) problem [BGV14, LS15] and its Hint-MLWE problem variant [KLSS23].

Definition 3 (MLWE). *Let k, ℓ, q be integers, and χ be a probability distribution over $\mathcal{R}$. The advantage of an adversary $\mathcal{A}$ against the MLWE problem $\text{MLWE}_{q, k, \ell, \chi}$ is defined as:*

$$\text{Adv}_{q, k, \ell, \chi}^{\text{MLWE}}(\mathcal{A}, 1^\lambda) = \left| \Pr\left[\mathcal{A}\left(A, [I_k \mid A] \cdot \vec{x}\right) = 1\right] - \Pr\left[\mathcal{A}\left(A, \vec{b}\right) = 1\right] \right|$$

where $A \xleftarrow{\$} \mathcal{R}_q^{k \times \ell}$, $\vec{b} \xleftarrow{\$} \mathcal{R}_q^k$, and $\vec{x} \leftarrow \chi^{k+\ell}$. The MLWE assumption states that any efficient adversary $\mathcal{A}$ has a negligible advantage. We simply write $\text{MLWE}_{q, k, \ell, \sigma}$ when $\mathcal{D}$ is the discrete Gaussian distribution with parameter σ .

The Hint-MLWE problem [KLSS23] states that the adversary cannot distinguish the MLWE distribution from the uniform distribution, even when ‘linear hints’ about the secret are provided to the adversary. We describe a specific variant of Hint-MLWE below.

Definition 4 (Hint-MLWE). Let k, ℓ, q, Q integers and χ, χ' probability distributions over $\mathcal{R}$, and $(\vec{v}_i)_{i \in [Q]}$ vectors in $\mathcal{R}^{k+\ell}$. The advantage of an adversary $\mathcal{A}$ against the Hint-MLWE problem $\text{HMLWE}_{q,k,\ell,Q,\chi,\chi',(\vec{v}_i)_{i \in [Q]}}$ is defined as:

$$\begin{aligned} & \text{Adv}_{q,k,\ell,Q,\chi,\chi',(\vec{v}_i)_{i \in [Q]}}^{\text{HMLWE}}(\mathcal{A}) \\ &= \left| \Pr \left[\mathcal{A} \left(A, [I_k \mid A] \cdot \vec{x}, (\vec{v}_i, z_i)_{i \in [Q]} \right) \right] = 1 \right] - \Pr \left[\mathcal{A} \left(A, \vec{b}, (\vec{v}_i, z_i)_{i \in [Q]} \right) = 1 \right] \right| \end{aligned}$$

where $A \xleftarrow{\$} \mathcal{R}_q^{k \times \ell}$, $\vec{b} \xleftarrow{\$} \mathcal{R}_q^k$ and $\vec{x} \leftarrow \chi^{k+\ell}$. Moreover, $z_i = \vec{v}_i^\top \vec{x} + y_i$ where $y_i \leftarrow \chi'$. The $\text{HMLWE}_{q,k,\ell,Q,\chi,\chi',(\vec{v}_i)_{i \in [Q]}}$ assumption states that any efficient adversary $\mathcal{A}$ has a negligible advantage. We write $\text{HMLWE}_{q,k,\ell,Q,\sigma_{\vec{x}},\sigma_y,(\vec{v}_i)_{i \in [Q]}}$ as a shorthand when χ and χ' are discrete Gaussian distributions with parameters $\sigma_{\vec{x}}$ and σ_y , respectively.

In this work, we will only rely on a restrained version of the Hint-MLWE problem where the hint vectors $\vec{v}_i$ are standard basis vectors in $\mathcal{R}^{k+\ell}$, and there are Q copies of each standard basis vector. The following lemma directly follows from Theorem 1 in [KLSS23].

Lemma 4 (Hardness of HMLWE). Let ℓ, k, q, Q be integers, $\mathcal{E}$ be the tuple of $(k+\ell)Q$ vectors containing Q copies of each $(1, 0, \dots, 0), \dots, (0, 0, \dots, 1) \in \mathcal{R}^{k+\ell}$, and $\sigma_{\vec{x}}$ and σ_y be some positive reals.

Let $\sigma > 0$ defined as $\sigma^{-2} = 2(\sigma_{\vec{x}}^{-2} + Q\sigma_y^{-2})$. If $\sigma \geq \eta_\epsilon(\mathbb{Z}^N)/\sqrt{\pi}$ for some $0 < \epsilon \leq 1/2$, then there exists an efficient reduction from $\text{MLWE}_{q,k,\ell,\sigma}$ to $\text{HMLWE}_{q,k,\ell,(k+\ell)Q,\sigma_{\vec{x}},\sigma_y,\mathcal{E}}$ that reduces the advantage by at most $2(k+\ell)\epsilon$.

2.4 Definitions and Security Models for tPKE

In this section, we recall the syntax and security notions of Threshold Public-key Encryption (tPKE), as recently consolidated in [BKW26]. We focus on the *asynchronous* setting where the set of decryptors is not known at the time of partial decryptions.

Definition 5 (tPKE). A t -out-of- n tPKE scheme is a tuple of algorithms $\text{tPKE} = (\text{KeyGen}, \text{Enc}, \text{PartDec}, \text{FinDec})$ defined as follows, where $\mathcal{PT}, \mathcal{CT}$ and $\mathcal{RD}$ denote the plaintext, ciphertext, and encryption randomness spaces, respectively, and χ_{rnd} denotes the distribution over $\mathcal{RD}$ from which honest encryption draws its randomness:

KeyGen($1^\lambda, t, n$): Upon input the security parameter λ , threshold t , and number of parties n , outputs a public key pk and n shares $\text{sk}_1, \dots, \text{sk}_n$ of the secret key.

Enc(pk, m): Upon input a public key pk and a message $m \in \mathcal{PT}$, samples $\text{rnd} \xleftarrow{\$} \chi_{\text{rnd}}$ and returns a ciphertext $\text{ct} \in \mathcal{CT}$. We write $\text{Enc}(\text{pk}, m; \text{rnd})$ to denote the (deterministic) encryption process using the specific encryption randomness $\text{rnd} \in \mathcal{RD}$.

PartDec(sk_i, ct): Takes as input the i -th key share sk_i and a ciphertext $\text{ct} \in \mathcal{CT}$, and returns a partial decryption pd_i .

FinDec(ct, {pd_i}_{i ∈ I}): Takes as input ciphertext $\text{ct} \in \mathcal{CT}$ and partial decryptions $\{\text{pd}_i\}_{i \in \mathcal{I}}$ of some subset $\mathcal{I} \subseteq [n]$ of size t , and returns a plaintext $m \in \mathcal{PT} \cup \{\perp\}$.

Definition 6 (Correctness). A t -out-of- n tPKE scheme $\text{tPKE} = (\text{KeyGen}, \text{Enc}, \text{PartDec}, \text{FinDec})$ is correct if

$$\Pr \left[\begin{array}{l} \text{ct} \leftarrow \text{Enc}(\text{pk}, m) \\ m' \neq m : \text{pd}_i \leftarrow \text{PartDec}(\text{sk}_i, \text{ct}) \text{ for } i \in \mathcal{I} \\ m' \leftarrow \text{FinDec}(\text{ct}, \{\text{pd}_i\}_{i \in \mathcal{I}}) \end{array} \right] \leq \text{negl}(\lambda)$$

for every message $m \in \mathcal{PT}$ and every subset $\mathcal{I} \subseteq [n]$ of size t , where the probability is taken over the random coins used in **KeyGen**, **Enc**, and **PartDec**.

We say that a threshold PKE scheme is perfectly correct if the decryption success probability is 1.

We now define CPA and CCA security for tPKE. As discussed in the introduction, prior works considered various security models making trade-offs adapted to their construction. We already do not wish to rely on any trusted third party assumption, or hiding of the ciphertexts' randomness generated by the encryption oracle. It still remains several dimensions studied in [BKW26]: the type of security (indistinguishability vs simulation security), number of corruptions supported ($= t-1$ or $\leq t-1$), support of partial decryption of challenge ciphertexts, support of *honest* or *semi-malicious* ciphertexts, *static* or *adaptive* corruption of parties.

We aim for the strongest security in all dimensions as introduced in [BKW26], at the exception of *adaptive* corruptions, which appears particularly challenging to achieve with a polynomial moduli. Namely, we will target CPA and CCA simulation security, under a *semi-malicious* encryption model. We will support any $\leq t-1$ corruptions, and all oracles proposed in [BKW26].

Semi-malicious Simulation Security. In this model, the adversary may supply any encryption randomness within the valid space—it need not follow the scheme's prescribed distribution—and security is formalized via a simulator rather than mere indistinguishability. The simulator must produce the public key and the corrupt parties' secret key shares without access to honest parties' keys, and must faithfully simulate all partial decryptions seen by the adversary. This captures the general scenario where the encryptor is one of the adversarial parties or colludes with them, since the public key is public and any party can produce ciphertexts with arbitrary randomness.

In the Chosen-Ciphertext Attacks (CCA) game, the key difference is that the adversary can submit any ciphertext directly to the partial decryption oracle, without the restriction that it must be generated through EncO or ChalO.

Definition 7 (Semi-malicious CPA and CCA Security). A threshold PKE scheme tPKE is simulation secure against chosen plaintext attacks (CPA) in the presence of a semi-malicious encryptor, denoted SemiMalSimCPA security, (resp. chosen ciphertext attacks (CCA), denoted SimCCA security) if for any PPT adversary $\mathcal{A}$, there exists a PPT simulator $\mathcal{S}$ such that the advantage

$$\begin{aligned} \text{Adv}_{\text{tPKE}}^{\text{SemiMalSimCPA}}(\mathcal{A}, \mathcal{S}) &:= \left| \Pr \left[\text{SemiMalSimCPA}_{\text{tPKE}, \mathcal{A}, \mathcal{S}}^0(1^\lambda) = 1 \right] \right. \\ &\quad \left. - \Pr \left[\text{SemiMalSimCPA}_{\text{tPKE}, \mathcal{A}, \mathcal{S}}^1(1^\lambda) = 1 \right] \right| \\ (\text{resp. } \text{Adv}_{\text{tPKE}}^{\text{SimCCA}}(\mathcal{A}, \mathcal{S}) &:= \left| \Pr \left[\text{SimCCA}_{\text{tPKE}, \mathcal{A}, \mathcal{S}}^0(1^\lambda) = 1 \right] \right. \\ &\quad \left. - \Pr \left[\text{SimCCA}_{\text{tPKE}, \mathcal{A}, \mathcal{S}}^1(1^\lambda) = 1 \right] \right|) \end{aligned}$$

is negligible, where the security experiment is defined in Fig. 1 (resp. the same figure, dropping the boxed elements).

$\text{SimCCA}_{\text{tPKE}, \mathcal{A}, \mathcal{S}}^b(1^\lambda)$	$\text{ChalO}(m)$
$\boxed{\text{SemiMalSimCPA}_{\text{tPKE}, \mathcal{A}, \mathcal{S}}^b(1^\lambda)}$	
1 : $(t, n, \mathcal{C}, \text{st}_{\mathcal{A}}) \xleftarrow{\$} \mathcal{A}(1^\lambda)$ // Corrupt set $\mathcal{C}$ 2 : require $\mathcal{C} \subset [n] \wedge \mathcal{C} < t$ 3 : if $b = 0$ then $(\text{pk}, (\text{sk}_j)_{j \in [n]}) \xleftarrow{\$} \text{KeyGen}(1^\lambda, t, n)$ 4 : else $(\text{pk}, (\text{sk}_j)_{j \in \mathcal{C}}, \text{st}_{\mathcal{S}}) \xleftarrow{\$} \mathcal{S}(1^\lambda, t, n, \mathcal{C})$ 5 : $L_{\text{Enc}} \leftarrow \emptyset; L_{\text{Chal}} \leftarrow \emptyset; P[\cdot] \leftarrow \emptyset; M[\cdot] \leftarrow \emptyset$ 6 : $\text{st}_{\mathcal{A}} \leftarrow (\text{pk}, (\text{sk}_j)_{j \in \mathcal{C}}, \text{st}_{\mathcal{A}})$ 7 : return $b' \xleftarrow{\$} \mathcal{A}(\boxed{\text{EncO}}_{\text{ChalO}, \text{PartDecO}}(\text{st}_{\mathcal{A}}))$ $\text{EncO}(m, \text{rnd})$	1 : if $b = 0$ then $\text{ct} \xleftarrow{\$} \text{Enc}(\text{pk}, m)$ 2 : else $(\text{ct}, \text{st}_{\mathcal{S}}) \xleftarrow{\$} \mathcal{S}(\text{st}_{\mathcal{S}})$ 3 : $L_{\text{Chal}} \leftarrow L_{\text{Chal}} \cup \{\text{ct}\}; M[\text{ct}] \leftarrow m$ 4 : return ct $\text{PartDecO}(\text{ct}, j)$
1 : require $\text{rnd} \in \mathcal{RD}$ 2 : $\text{ct} \leftarrow \text{Enc}(\text{pk}, m; \text{rnd})$ 3 : if $b = 1$ then $\text{st}_{\mathcal{S}} \xleftarrow{\$} \mathcal{S}(m, \text{rnd}, \text{st}_{\mathcal{S}})$ 4 : $L_{\text{Enc}} \leftarrow L_{\text{Enc}} \cup \{\text{ct}\}; M[\text{ct}] \leftarrow m$ 5 : return ct	1 : require $\text{ct} \in L_{\text{Enc}} \cup L_{\text{Chal}}$ 2 : $P[\text{ct}] \leftarrow P[\text{ct}] \cup \{j\}$ 3 : if $b = 0$ then $\text{pd}_j \xleftarrow{\$} \text{PartDec}(\text{sk}_j, \text{ct})$ 4 : else 5 : if $ P[\text{ct}] \cup \mathcal{C} \geq t \wedge \text{ct} \in L_{\text{Chal}}$ then 6 : $m \leftarrow M[\text{ct}]; (\text{pd}_j, \text{st}_{\mathcal{S}}) \xleftarrow{\$} \mathcal{S}(\text{ct}, m, j, \text{st}_{\mathcal{S}})$ 7 : else $(\text{pd}_j, \text{st}_{\mathcal{S}}) \xleftarrow{\$} \mathcal{S}(\text{ct}, j, \text{st}_{\mathcal{S}})$ 8 : return pd_j

Fig. 1. Simulation-based semi-malicious CPA and CCA security experiments for tPKE . The boxed requirement in PartDecO is part of $\text{SemiMalSimCPA}_{\text{tPKE}, \mathcal{A}, \mathcal{S}}^b$ only.

2.5 Shamir Secret Sharing

Secret sharings are useful primitives for threshold constructions. They distribute a secret among a set of participants, such that the secret can be reconstructed

by a sufficiently large subset of parties (typically any subset of at least t parties). Among the various secret sharing schemes, Shamir Secret Sharing (SSS) is particularly appealing due to its simplicity and small share size. We consider below that the secret shared is stored at the evaluation point 0.

Definition 8 (Shamir Secret Sharing). *Let $\mathcal{P} = [n]$ be a set of participants, $t \leq n$ be a threshold, and $\mathcal{S} = \{\alpha_1, \dots, \alpha_n\} \subset \mathcal{R}_q$ be evaluation points such that $-\alpha_j$ and $\alpha_i - \alpha_j$ is invertible in $\mathcal{R}_q$ for all distinct $0 \leq i, j \leq n$. For $\mathcal{I} \subseteq [n]$ and $i \in \mathcal{I}$, the Lagrange coefficient is*

$$\lambda_i^{\mathcal{I}} := \prod_{j \neq i \in \mathcal{I}} (-\alpha_j) \cdot (\alpha_i - \alpha_j)^{-1} \bmod q.$$

Then, t -out-of- n threshold Shamir Secret Sharing over $\mathcal{R}_q$ consists of the following two algorithms.

- **SS.Share _{t,n} (v):** *Upon input a value $v \in \mathcal{R}_q$, the dealer samples $r_j \leftarrow \mathcal{R}_q$ for $1 \leq j < t$, and computes $s_i = v + \sum_{j=1}^{t-1} r_j \cdot \alpha_i^j \bmod q$. Output $\{(s_i, \alpha_i)\}_{i \in [n]}$.*
- **SS.Combine _{t,n} ($\{s_i\}_{i \in \mathcal{I}}$):** *Upon input secret shares $\{s_i\}_{i \in \mathcal{I}}$ for any subset $\mathcal{I} \subset [n]$ such that $|\mathcal{I}| = t$, compute and output $s = \sum_{i \in \mathcal{I}} \lambda_i^{\mathcal{I}} \cdot s_i \bmod q$*

We extend this notation to the secret sharing of vectors as well. More precisely, for a vector $\vec{v} = (v_1, \dots, v_k) \in \mathcal{R}_q^k$, the notation **SS.Share _{t,n} ($\vec{v}$)** means applying **SS.Share _{t,n}** component-wise to each coordinate of $\vec{v}$.

Now we discuss how Shamir Secret Sharing can be leveraged in a thresholded Regev-style encryption scheme. Let $\vec{s}_1, \dots, \vec{s}_n$ be the shares of the secret vector $\vec{s} \in R^k$, i.e., **SS.Share _{t,n} ($\vec{v}$)** = $\{(\vec{s}_i, \alpha_i)\}_{i \in [n]}$. Then, for some user set $\mathcal{I} \subseteq [n]$ to decrypt an MLWE ciphertext $(b, \vec{a}) \in \mathcal{R}_q^{k+1}$, each i -th party outputs the partial decryption $d_i := \langle \vec{a}, \vec{s}_i \rangle + f_i \bmod q$ for some flooding noise f_i .

Upon receiving the partial decryptions $\{d_i\}_{i \in \mathcal{I}}$, we wish to leverage Lagrange interpolation to combine them. Naively, one would compute $b + \sum_{i \in \mathcal{I}} \lambda_i^{\mathcal{I}} d_i \bmod q$, which would yield $b + \langle \vec{a}, \vec{s} \rangle + \sum_{i \in \mathcal{I}} \lambda_i^{\mathcal{I}} f_i$. However, the Lagrange coefficients $\lambda_i^{\mathcal{I}}$ can have large norms, which would amplify the flooding noise $\sum_{i \in \mathcal{I}} \lambda_i^{\mathcal{I}} f_i$ and prevent correct decryption. To address this issue, prior works have introduced a clearing factor γ during decryption that ensures that the scaled Lagrange coefficients have small norms. Thus, parties compute $\gamma b + \sum_{i \in \mathcal{I}} \gamma \lambda_i^{\mathcal{I}} d_i \bmod q$, which becomes $\gamma \cdot (b + \langle \vec{a}, \vec{s} \rangle \bmod q) + \sum_{i \in \mathcal{I}} \gamma \lambda_i^{\mathcal{I}} f_i$ and allows correct decryption as long as $\gamma \lambda_i^{\mathcal{I}} f_i$ is small enough for all $i \in \mathcal{I}$.

In order to minimize the size of γ and the $\lambda_i^{\mathcal{I}}$ it is crucial to choose the α_i carefully. Selecting monomials $\alpha_i = X^i$ has so far provided the most compact parameters for this purpose [BFM⁺25, CLW25, KLL⁺26]. We adopt this choice and characterize it formally in Lem. 5.

Lemma 5 ([KLL⁺26, Thm 4.3]). *Assume q is odd. The monomial set $\mathcal{S} = \{\alpha_i = X^{i-1}\}_{1 \leq i \leq 2N}$ over $\mathcal{R}$ satisfies the invertibility property of Def. 8, and for the clearing factor $\gamma = 2^{\lceil \log t \rceil}$, we define the bound B_λ and $B_{\lambda^{-1}}$ as some*

integers such that $\|\gamma\lambda_i^{\mathcal{I}}\|_1 \leq B_\lambda$ and $\|(\lambda_i^{\mathcal{I}})^{-1}\|_1 \leq B_{\lambda^{-1}}$ for all $i \in \mathcal{I}$. Then, the following holds for any $\mathcal{I} \subseteq_t [n]$.

- $B_\lambda \leq 2^{-t+\lceil \log t \rceil+1} \cdot \sqrt{N} \cdot \prod_{1 \leq k \leq \lfloor t/2 \rfloor} \sin^{-2}(\pi \frac{k}{2N})$
- $B_{\lambda^{-1}} \leq 2^{t-1} \cdot \sqrt{N} \cdot \prod_{1 \leq k \leq \lfloor t/2 \rfloor-1} \cos^2(\pi \frac{k}{2N})$

3 Adaptive Hint-MLWE

In this section, we introduce Adaptive Hint-MLWE, a variant of Hint-MLWE in which the adversary can choose hint coefficients adaptively, and we provide a reduction to MLWE. Adaptive Hint-MLWE is a natural extension of the Hint-MLWE problem, but previous reduction techniques do not extend to this adaptive setting. We believe the problem is of independent interest and may be more broadly useful. In Def. 9 we give a formal definition of Adaptive Hint-MLWE.

Definition 9 (Adaptive Hint-MLWE). Let k, ℓ, q, Q be integers, χ, χ' be probability distributions over $\mathcal{R}$, and $\mathcal{V}$ be a set over $\mathcal{R}^{k+\ell}$. The advantage of an adversary $\mathcal{A}$ making at most Q queries to the adaptive hint oracle $\mathcal{O}_{\text{hint}}$ against the Adaptive Hint-MLWE problem $\text{AHMLWE}_{q,k,\ell,Q,\chi,\chi',\mathcal{V}}$ is defined as:

$$\text{Adv}_{q,k,\ell,Q,\chi,\chi',\mathcal{V}}^{\text{AHMLWE}}(\mathcal{A}) = \left| \Pr[\text{Game}^{\text{AHMLWE},0}(\mathcal{A}) = 1] - \Pr[\text{Game}^{\text{AHMLWE},1}(\mathcal{A}) = 1] \right|$$

where $\text{Game}^{\text{AHMLWE},b}(\mathcal{A})$ is shown in Fig. 2. The $\text{AHMLWE}_{q,k,\ell,Q,\chi,\chi',\mathcal{V}}$ assumption states that any efficient adversary $\mathcal{A}$ has only negligible advantage. We may write $\text{AHMLWE}_{q,k,\ell,Q,\sigma_{\vec{x}},\sigma_y,\mathcal{V}}$ as shorthand when χ and χ' are discrete Gaussian distributions with parameters $\sigma_{\vec{x}}$ and σ_y , respectively. We refer to the case $b = 0$ as the ‘Random’ case, and the case $b = 1$ as the ‘Real’ case.

$\text{Game}^{\text{AHMLWE},b}(\mathcal{A})$	$\mathcal{O}_{\text{hint}}(\vec{v})$
1 : $\text{ctr} := 0$	1 : require $\vec{v} \in \mathcal{V} \wedge \text{ctr} < Q$
2 : $A \leftarrow \mathcal{R}_q^{k \times \ell}, \vec{x} \leftarrow \chi^{k+\ell}$	2 : $\text{ctr} \leftarrow \text{ctr} + 1$
3 : if $b = 0$ then $\vec{b} \leftarrow \mathcal{R}_q^k$	3 : $y \leftarrow \chi'$
4 : if $b = 1$ then $\vec{b} := [I_k \mid A] \cdot \vec{x}$	4 : return $\vec{v}^\top \vec{x} + y$
5 : return $\mathcal{A}^{\mathcal{O}_{\text{hint}}}(A, \vec{b})$	

Fig. 2. Security experiments for the AHMLWE problem.

To prove the hardness of Adaptive Hint-MLWE, we adapt the rerandomization lemma from [KY16] into the discrete Gaussian setting. Roughly speaking, the rerandomization lemma allows one to take selective hints and process them to generate adaptive hints on-the-fly. To enable this, we first introduce some useful lemmas.

Lemma 6 (Simplified convolution Lemma, adapted from [dCKK⁺26]).

Let $\mathbf{M} \in \mathbb{Z}^{m \times n}$ be a matrix, $\sigma > 0$ be a real, and $\Sigma \succeq 0$ be a matrix such that

$$\frac{1}{\sigma^2} + \|\mathbf{M}\sqrt{\Sigma^{-1}}\|_2^2 \leq \frac{2\pi}{\eta_\varepsilon(\mathbb{Z}^n)^2}$$

for some $0 < \varepsilon \leq 1/2$. Then, the following distribution over $\mathbb{Z}^m$

$$\{\mathbf{M}\mathbf{e} + \mathbf{f} \mid \mathbf{e} \leftarrow D_{\mathbb{Z}^n, \sigma}, \mathbf{f} \leftarrow D_{\mathbb{Z}^m, \sqrt{\Sigma}}\}$$

is within statistical distance 4ε of $D_{\mathbb{Z}^m, \sqrt{\sigma^2 \mathbf{M}\mathbf{M}^\top + \Sigma}}$.

Proof. See Sec. A.

Lemma 7 (Gaussian rerandomization, adapted from [KY16]). Let q, ℓ, m be positive integers. Let $\mathbf{x} \in \mathbb{Z}^n$ be an arbitrary vector and let $\mathbf{y}$ be sampled from $D_{\mathbb{Z}^n, \sigma}$. Then, there exists a PPT algorithm $\text{Rerand}(\mathbf{M}, \mathbf{x} + \mathbf{y}, \sigma, s)$ that outputs $\mathbf{M}\mathbf{x} + \mathbf{y}'$ such that the distribution of $\mathbf{y}'$ is within statistical distance 4ε of $D_{\mathbb{Z}^m, \sigma s}$, if $\sigma > \eta_\varepsilon(\mathbb{Z}^n)/\sqrt{2\pi}$ and $\|\mathbf{M}\|_2 \leq \sqrt{1 - \eta_\varepsilon^2(\mathbb{Z}^n)/(2\pi\sigma^2)} \cdot s$.

Proof. See Sec. A.

We show that the Adaptive Hint-MLWE problem is no easier than the MLWE problem. The crux of the argument is to rely on the noise-rerandomization technique from [KY16].

Theorem 1 (Hardness of AHMLWE). For any integers ℓ, k, q, N, Q , set $\mathcal{V} \subset \mathcal{R}^{k+\ell}$, and positive reals $B_{\text{hint}}, \sigma, \sigma_{\bar{x}}, \sigma_y$ such that

$$\begin{aligned} & - \sum_{0 \leq j < k+\ell} \|v_i\|_1^2 \leq B_{\text{hint}} \text{ for any } \vec{v} = (v_i)_{0 \leq i < k+\ell} \in \mathcal{V}, \\ & - \sigma_y \geq \sqrt{2B_{\text{hint}}/\pi} \cdot \eta_\varepsilon(\mathbb{Z}^N) \\ & - \sigma \geq \eta_\varepsilon(\mathbb{Z}^N)/\sqrt{\pi} \\ & - \\ & \frac{1}{\sigma^2} = 2 \left(\frac{1}{\sigma_{\bar{x}}^2} + \frac{\kappa^2 Q}{\sigma_y^2} \right) \text{ for } \kappa := \sqrt{\frac{1 - \sqrt{1 - 2B_{\text{hint}}\eta_\varepsilon^2(\mathbb{Z}^N)/(\pi\sigma_y^2)}}{\eta_\varepsilon^2(\mathbb{Z}^N)/(\pi\sigma_y^2)}}. \end{aligned}$$

Then, the $\text{AHMLWE}_{q,k,\ell,Q,\sigma_{\bar{x}},\sigma_y,\mathcal{V}}$ problem is as hard as the $\text{MLWE}_{q,k,\ell,\sigma}$ problem, with an advantage loss at most $(4Q + 2(k+\ell))\varepsilon$.

Proof. Let $\mathcal{E}$ be the set of $(k+\ell)Q$ vectors containing Q copies of each $(1, 0, \dots, 0), \dots, (0, 0, \dots, 1) \in \mathcal{R}^{k+\ell}$. Our proof strategy consists of constructing an adversary $\mathcal{B}$ against the $\text{HMLWE}_{q,k,\ell,(k+\ell)Q,\sigma_y',\mathcal{E}}$ problem from an adversary $\mathcal{A}$ against the problem $\text{AHMLWE}_{q,k,\ell,Q,\sigma_{\bar{x}},\sigma_y,\mathcal{V}}$ by simulating adaptive hints from selective hints $x_i + y_i$ for $y_i \leftarrow D_{\sigma_y'}$ for some positive real σ_y' to be determined as a function of σ_y . We achieve this by using the rerandomization technique from [KY16], which provides an algorithm $\text{Rerand} : (\mathbf{M}, \mathbf{x} + \mathbf{y}) \mapsto \mathbf{M}\mathbf{x} + \mathbf{y}'$. We then rely on the reduction between $\text{HMLWE}_{q,k,\ell,(k+\ell)Q,\sigma_{\bar{x}},\sigma_y',\mathcal{E}}$ and $\text{MLWE}_{q,k,\ell,\sigma}$ from Lem. 4.

Let us start by reducing $\text{AHMLWE}_{q,k,\ell,Q,\sigma_{\vec{x}},\sigma_y,\mathcal{V}}$ to $\text{HMLWE}_{q,k,\ell,(k+\ell)Q,\sigma_{\vec{x}},\sigma'_y,\mathcal{E}}$. Consider an adversary $\mathcal{A}$ against the $\text{AHMLWE}_{q,k,\ell,Q,\sigma_{\vec{x}},\sigma_y,\mathcal{V}}$ problem. Next, let us define an adversary $\mathcal{B}$ against the $\text{HMLWE}_{q,k,\ell,(k+\ell)Q,\sigma_{\vec{x}},\sigma'_y:=\sigma_y/\kappa,\mathcal{E}}$ problem with hints $z_{i,j} := x_j + y_{i,j}$ for $i \in [Q]$ and $0 \leq j < k + \ell$ where $y_{i,j} \leftarrow D'_{\sigma_y}$. Now we simulate each i -th call $\mathcal{O}_{\text{hint}}(\mathcal{V})$ by returning a polynomial where the coefficient vector is the output of $\text{Rerand}(\mathbf{V}_i, \mathbf{z}_i, \sigma'_y, s, \delta)$ where $\mathbf{V}_i$ is a concatenation of negacyclic matrices $\mathbf{V}_{i,j}$ corresponding to the j -th entry $v_{i,j}$ of the query vector $\vec{v}_i$ for $0 \leq j < k + \ell$, and $\mathbf{z}_i$ is a concatenation of coefficient vectors of each $z_{i,0}, \dots, z_{i,k+\ell-1}$. We note that $\mathbf{V}_i \cdot \mathbf{z}_i$ is essentially the coefficient vector of the ring element $\vec{v}_i^\top \cdot (z_{i,j})_{0 \leq j < k+\ell}$.

We observe that

$$\begin{aligned} \|\mathbf{V}_i\|_2^2 &= \left\| \sum_{j=0}^{k+\ell-1} \mathbf{V}_{i,j}^\top \mathbf{V}_{i,j} \right\|_2 \leq \sum_{j=0}^{k+\ell-1} \|\mathbf{V}_{i,j}^\top \mathbf{V}_{i,j}\|_2 \leq \sum_{j=0}^{k+\ell-1} \|v_{i,j}\|_1^2 \\ &\leq B_{\text{hint}} = \kappa^2 \left(1 - \frac{\eta_\varepsilon^2(\mathbb{Z}^N) \kappa^2}{2\pi\sigma_y^2} \right) \end{aligned}$$

where the last equality is derived from the definition of κ . On the other hand, $\sigma'_y > \eta_\varepsilon(\mathbb{Z}^N)/\sqrt{2\pi}$ since $2\sqrt{B_{\text{hint}}} > \kappa$, by definition of κ as well.

Therefore, Lem. 7 applies, and the simulated $\mathcal{O}_{\text{hint}}$ outputs a vector $\mathbf{V}_i \mathbf{x} + \mathbf{y}'_i$ where $\mathbf{x}$ is a concatenation of the coefficient vectors of the secret $\vec{x}$ and the distribution of $\mathbf{y}'_i$ is within statistical distance 4ε from $D_{\mathbb{Z}^N, \sigma_y}$. We remark that this can be seen as a coefficient vector of $\vec{v}_i^\top \cdot \vec{x} + y'_i$ for some ring element y'_i having $\mathbf{y}'_i$ as the coefficient vector. Therefore, $\mathcal{B}$ simulates the view of $\mathcal{A}$ up to statistical distance $Q \cdot 4\varepsilon$, and we conclude that if $\mathcal{A}$ breaks $\text{AHMLWE}_{q,k,\ell,Q,\sigma_{\vec{x}},\sigma_y,\mathcal{V}}$, then $\mathcal{B}$ breaks $\text{HMLWE}_{q,k,\ell,(k+\ell)Q,\sigma_{\vec{x}},\sigma'_y,\mathcal{E}}$.

Now we reduce $\text{HMLWE}_{q,k,\ell,(k+\ell)Q,\sigma_{\vec{x}},\sigma'_y,\mathcal{E}}$ to $\text{MLWE}_{q,k,\ell,\sigma}$. This directly follows from Lem. 4, since

$$\frac{1}{\sigma^2} = 2 \left(\frac{1}{\sigma_{\vec{x}}^2} + \frac{\kappa^2 Q}{\sigma_y^2} \right) = 2 \left(\frac{1}{\sigma_{\vec{x}}^2} + \frac{Q}{\sigma_y'^2} \right),$$

as all the vectors $e \in \mathcal{E}$ satisfy $\|e\|_1 = 1$. Therefore, the overall advantage loss from Hint-MLWE to MLWE is $(4Q + 2(k + \ell))\varepsilon$, as claimed. $\square$

4 Our Threshold PKE Scheme

In this section, we present our tPKE scheme and analyze its correctness and security. Our construction can be viewed as a direct thresholdization of the standard MLWE-based PKE scheme, which also serves as the underlying CPA-secure component of CRYSTALS-Kyber [BDK⁺18], combined with Shamir secret sharing and a standard one-round distributed decryption protocol. To facilitate our security reduction, we introduce minor modifications to this standard template, specifically employing discrete Gaussian distributions for key and encryption randomness generation.

Although this design strategy is conceptually natural and similar approaches have been adopted in several prior works, formally proving its security under standard assumptions has remained an open challenge as we discussed in the introduction. This work is the first to formally prove the strongest form of CPA security of a direct thresholdization of the standard MLWE-based PKE scheme under standard assumptions.

4.1 Scheme Description

The construction of our tPKE scheme is given in Figure 3. We assume that the ciphertext modulus q is odd, ensuring in particular that $\gamma^{-1} \bmod q$ exists.

To enable asynchronous decryption, we perform the final recombination in FinDec after multiplying by γ , namely by adding γc_0 and scaling each partial decryption by γ times its corresponding Lagrange coefficient; the point of this factor is that, for the evaluation points we use, these effective coefficients remain small. This normalization trick is standard in this setting and already appears, for instance, in Pilvi [CLW25]. Accordingly, during encryption we compensate for this extra factor by multiplying the message by γ^{-1} . As a result, the original message is correctly reconstructed after final decryption. Crucially, this design allows the partial decryption algorithm to operate without knowing the decryption set in advance, as it does not require the Lagrange coefficients.

Moreover, we make partial decryption deterministic by deriving the flooding term from a PRF keyed by the party's secret state. This removes the need to restrict each party to at most one partial decryption per ciphertext, which Pilvi avoided by imposing exactly such a limitation.

4.2 Correctness and Security

In this section, we analyze the correctness and prove the security of our tPKE scheme. We will rely on the hardness of the Adaptive Hint-MLWE problem.

Theorem 2 (Correctness). *Assuming F is a secure PRF, the tPKE scheme described above is correct, except with negligible probability, if the following holds:*

$$\gamma \left(\rho_r \sigma_r + e^{1/2} \cdot 2kN \sigma_r \sigma_s \right) + tB_\lambda \rho_f \sigma_f < \frac{q}{2p} - \frac{p}{2},$$

where, for the target (negligible) failure probability ε and $\rho(\varepsilon, \cdot)$ as in Lem. 2, $\rho_r := \rho(\varepsilon, 1)$ and $\rho_f := \rho(\varepsilon, t)$, and B_λ is as defined in Lem. 5.

Proof. By the correctness of the baseline Shamir Secret Sharing scheme, the (final) phase of public key encryption is

$$\begin{aligned} \gamma b + \sum_{i \in \mathcal{I}} \gamma \lambda_i^T \cdot d_i &= \gamma \left(\gamma^{-1} \cdot \left\lfloor \frac{q}{p} \right\rfloor m + r_0 + \vec{r}_1^T \vec{s} + \vec{r}_2^T \vec{e} \right) + \sum_{i \in \mathcal{I}} \gamma \lambda_i^T f_i \pmod{q} \\ &= \left\lfloor \frac{q}{p} \right\rfloor m + \gamma(r_0 + \vec{r}_1^T \vec{s} + \vec{r}_2^T \vec{e}) + \sum_{i \in \mathcal{I}} \gamma \lambda_i^T f_i \pmod{q}. \end{aligned}$$

Threshold PKE

Given a Shamir Secret Sharing scheme $\text{SS} = (\text{Share}, \text{Combine})$, we describe a CPA-secure threshold PKE scheme tPKE as follows.

- **KeyGen**($1^\lambda, t, n$): Upon input the security parameter λ , threshold t , and number of parties n , sample the secret $\vec{s} \leftarrow D_{\sigma_s}^k$, and let $\{(\vec{s}_i, \alpha_i)\}_{i \in [n]} \leftarrow \text{SS.Share}_{t,n}(\vec{s})$. Sample $A \leftarrow \mathcal{U}(\mathcal{R}_q^{k \times k})$ and $\vec{e} \leftarrow D_{\sigma_e}^k$, and let $\vec{b} = -A\vec{s} + \vec{e} \pmod{q}$. For $1 \leq i \leq n$, sample a PRF key K_i and send the public key $\text{pk} = (\vec{b}, A) \in \mathcal{R}_q^{k \times (k+1)}$ and a key share $(\vec{s}_i, \alpha_i, K_i)$ to the i -th party.
- **Enc**(pk, m): Upon input a public key $\text{pk} = (\vec{b}, A)$, and a message $m \in \mathcal{R}_p$, sample $r_0 \leftarrow D_{\sigma_r}$, $\vec{r}_1 \leftarrow D_{\sigma_r}^k$ and $\vec{r}_2 \leftarrow D_{\sigma_r}^k$. Compute $c_0 = \vec{r}_2^\top \vec{b} + r_0 + \gamma^{-1} \cdot \lfloor \frac{q}{p} \rfloor m \pmod{q}$ and $\vec{c}_1 = A^\top \vec{r}_2 + \vec{r}_1 \pmod{q}$, and output $\text{ct} = (c_0, \vec{c}_1)$.
- **PartDec**(s_i, ct): Upon input a secret key share $s_i = (\vec{s}_i, \alpha_i, K_i)$, and a ciphertext $\text{ct} = (c_0, \vec{c}_1) \in \mathcal{R}_q^{k+1}$, compute $f_{\text{ct},i} := F(K_i, \text{ct})$ and output $d_{\text{ct},i} = \vec{c}_1^\top \vec{s}_i + f_{\text{ct},i} \pmod{q}$. Here, $F : \text{Keys} \times \mathcal{R}_q^{k+1} \rightarrow \mathcal{R}$ is a deterministic PRF whose output, for a uniformly random key, is required to be D_{σ_f} -distributed rather than uniform. This makes $f_{\text{ct},i}$ fixed and reproducible for each (s_i, ct) , so repeated partial decryptions of the same ciphertext by the same party remain consistent.
- **FinDec**($\text{ct}, \{d_i\}_{i \in \mathcal{I}}, \mathcal{I}$): Upon input a ciphertext $\text{ct} = (c_0, \vec{c}_1)$ and partial decryptions $\{d_i\}_{i \in \mathcal{I}}$ for some subset $\mathcal{I} \subseteq [n]$, compute and output the following:

$$m = \left\lfloor \frac{p}{q} \left(\gamma c_0 + \sum_{i \in \mathcal{I}} \gamma \lambda_i^\top d_i \pmod{q} \right) \right\rfloor$$

Fig. 3. Our tPKE construction.

For this ciphertext to be decrypted correctly, the infinite norm of the total error $e_{\text{ct}} := \gamma(r_0 + \vec{r}_1^\top \vec{s} + \vec{r}_2^\top \vec{e}) + \sum_{i \in \mathcal{I}} \gamma \lambda_i^\top f_i$ needs to be smaller than $\frac{q}{2p} - \frac{1}{2}$.

Since each $f_i = F(K_i, \text{ct})$ is derived from a PRF key never revealed to anyone, the hybrid argument of **Game**₁ in the proof of Thm. 3 shows that $(f_i)_{i \in \mathcal{I}}$ is, up to advantage $t \cdot \text{Adv}_F^{\text{PRF}}(\mathcal{D})$, distributed as t independent samples from D_{σ_f} , which we assume henceforth.

By Lem. 1 applied to $\vec{u} = (\vec{r}_1, \vec{r}_2)$ and $\vec{v} = (\vec{s}, \vec{e})$, $\|\vec{r}_1^\top \vec{s} + \vec{r}_2^\top \vec{e}\|_\infty \leq \|(\vec{r}_1, \vec{r}_2)\|_2 \cdot \|(\vec{s}, \vec{e})\|_2$. By Lem. 3 (applied with $\ell = 2k$ to $(\vec{r}_1, \vec{r}_2) \leftarrow D_{\sigma_r}^{2k}$ and to $(\vec{s}, \vec{e}) \leftarrow D_{\sigma_s}^{2k}$, each except with probability at most $2^{-2kN/10}$), this is in turn bounded by $e^{1/4} \sigma_r \sqrt{2kN} \cdot e^{1/4} \sigma_s \sqrt{2kN} = e^{1/2} \cdot 2kN \sigma_r \sigma_s$.

By Lem. 2 (applied with $\ell = 1$ to $r_0 \leftarrow D_{\sigma_r}$, and with $\ell = t$ to $(f_i)_{i \in \mathcal{I}} \leftarrow D_{\sigma_f}^t$, each except with negligible probability) and Lem. 5, $\|r_0\|_\infty \leq \rho_r \sigma_r$ and $\|\sum_{i \in \mathcal{I}} \gamma \lambda_i^\top f_{\text{ct},i}\|_\infty \leq t B_\lambda \rho_f \sigma_f$.

Combining these bounds via the triangle inequality, except with negligible probability over the union of the above events (and the PRF advantage), $\|e_{\text{ct}}\|_\infty \leq \gamma(\rho_r \sigma_r + e^{1/2} \cdot 2kN \sigma_r \sigma_s) + t B_\lambda \rho_f \sigma_f \leq \frac{q}{2p} - \frac{p}{2}$ from the conditions. $\square$

Theorem 3 (Security). *The tPKE scheme in Fig.3 is semi-malicious simulation secure against CPA attacks under the security of the PRF F , and the hardness of the problems $\text{AHMLWE}_{q,k,k,Q,\sigma_s,\sigma_f,\mathcal{V}_{B_1}}$, $\text{AHMLWE}_{q,k+1,k,n,\sigma_r,\sigma_f,\mathcal{V}_{B_2}}$, and $\text{AHMLWE}_{q,k+Q_c,k,Q,\sigma_1,\sigma_f,\mathcal{V}_{B_3}}$ assumptions, for $\sigma_1^2 = \frac{\sigma_f^2}{2\gamma^2} \left(1 + \sqrt{1 - \frac{2\gamma^2\eta_\varepsilon(\mathbb{Z}^N)^2}{\pi\sigma_f^2}}\right)$.*

Formally, if there exists an adversary $\mathcal{A}$ against the SemiMalSimCPA game making at most Q queries to the partial decryption oracle PartDecO and Q_c queries to the challenge oracle ChalO, and corrupting a set $\mathcal{C}$ of parties, there exists a simulator $\mathcal{S}$, a PRF distinguisher $\mathcal{D}$, and AHMLWE adversaries $\mathcal{B}_1$, $\mathcal{B}_2$, and $\mathcal{B}_3$, such that

$$\begin{aligned} \text{Adv}_{\text{tPKE}}^{\text{SemiMalSimCPA}}(\mathcal{A}, \mathcal{S}) &\leq \text{Adv}_{q,k,k,Q,\sigma_s,\sigma_f,\mathcal{V}_{B_1}}^{\text{AHMLWE}}(\mathcal{B}_1) + Q_c \cdot \text{Adv}_{q,k+1,k,n,\sigma_r,\sigma_f,\mathcal{V}_{B_2}}^{\text{AHMLWE}}(\mathcal{B}_2) \\ &\quad + (t-1) \cdot \text{Adv}_{q,k+Q_c,k,Q,\sigma_1,\sigma_f,\mathcal{V}_{B_3}}^{\text{AHMLWE}}(\mathcal{B}_3) + n \cdot \text{Adv}_F^{\text{PRF}}(\mathcal{D}) \\ &\quad + Q_c \cdot 2^{-N(2k+1)/10} + 2^{-2kN/10} + 4(t-1)Q_c\varepsilon, \end{aligned}$$

where $B_1 = e^{1/2}N^2(2k+1)(B_{\lambda-1}\sigma_r)^2$, $B_2 = B_{\lambda-1}^2(1 + 2e^{1/2}kN^2\sigma_s^2)$, $B_3 = e^{1/2}N^2(2k+1)\sigma_r^2 \max(\gamma, B_{\lambda-1}B_\lambda)^2$, and $B_\lambda, B_{\lambda-1}$ are as defined in Lem. 5.

Proof. We use a sequence of games to move from the game corresponding to the real world ($b = 0$) to the ideal world ($b = 1$) of the SemiMalSimCPA security definition, where the challenger simulates the view of the adversary $\mathcal{A}$. We bound $\mathcal{A}$'s change in winning probability between each pair of adjacent games, writing $p_i := \Pr[\text{Game}_{i,\mathcal{A}}(1^\lambda) = 1]$ for its winning probability in Game_i .

Throughout the proof, we denote by $\mathcal{C} \subseteq [n]$ the set of corrupted parties, and by $\mathcal{C}' \subseteq [n]$ an extended set such that $\mathcal{C} \subseteq \mathcal{C}'$ and $|\mathcal{C}'| = t-1$, deterministically chosen. We also fix the valid randomness space $\mathcal{RD}$ for semi-malicious EncO queries to be the two-norm ball

$$\mathcal{RD} := \left\{ \vec{r} \in \mathcal{R}_q^{2k+1} : \|\vec{r}\|_2 \leq e^{1/4}\sigma_r\sqrt{N(2k+1)} \right\},$$

matching the typical magnitude of an honestly-sampled $\vec{r} \leftarrow D_{\sigma_r}^{2k+1}$ by Lem. 3.

Game₀: This is the case $b = 0$ of the simulation security game SemiMalSimCPA, corresponding to the real world. The public key and secret key shares are generated honestly. Responses to ChalO and EncO queries are generated according to the real encryption algorithm, and responses to PartDecO queries are generated according to the real partial decryption algorithm.

$$p_0 = \Pr[\text{SemiMalSimCPA}_{\text{tPKE},\mathcal{A},\mathcal{S}}^0(1^\lambda) = 1]$$

Game₁: For every honest party $i \notin \mathcal{C}$, we replace the PRF-derived flooding noise $f_{\text{ct},i} := F(K_i, \text{ct})$ with a value sampled directly from D_{σ_f} , once per queried pair (ct, i) and memoized after that. For $i \in \mathcal{C}$, $f_{\text{ct},i}$ continues to be computed honestly via $F(K_i, \text{ct})$.

Since K_i for $i \notin \mathcal{C}$ is never revealed, this swap is exactly what the security of F guarantees, applied one key at a time over the (at most) n keys; this matches Game_0 once complete:

$$|p_0 - p_1| \leq n \cdot \text{Adv}_F^{\text{PRF}}(\mathcal{D}),$$

for the PRF distinguisher $\mathcal{D}$ that simulates $\mathcal{A}$, answering PartDecO via its own oracle for the key currently being swapped and honestly otherwise.

Game₂: In this hybrid, we equivalently rewrite the partial decryption oracle PartDecO to explicit dependencies on the public key, on shares of the parties in $\mathcal{C}'$, and on the secret $\vec{s}$. In particular, in this hybrid, the shares of parties $i \notin \mathcal{C}'$ are no longer directly used.

First, note that all ciphertexts queried to the partial decryption oracle PartDecO must have been generated either by ChalO or by EncO . In particular, for any ciphertext $\text{ct} = (c_0, \vec{c}_1)$ queried to PartDecO , there exist $m \in \mathcal{R}_p$ and $\vec{r} = (r_0, \vec{r}_1, \vec{r}_2) \in \mathcal{R}_q^{2k+1}$ such that $c_0 = \vec{r}_2^\top \vec{b} + r_0 + \gamma^{-1} \cdot \lfloor \frac{q}{p} \rfloor m$ and $\vec{c}_1 = A^\top \vec{r}_2 + \vec{r}_1 \pmod{q}$. These can be stored by the challenger when responding to ChalO and EncO queries, and retrieved when PartDecO is queried with ct .

Also, recall that since the key shares are generated via Shamir Secret Sharing, any share $\vec{s}_i$ for $i \notin \mathcal{C}'$ can be computed from the secret $\vec{s}$ and the shares of the parties in $\mathcal{C}'$ via Lagrange interpolation. To lighten notation, for $i \notin \mathcal{C}'$ we write $\mu_i := (\lambda_i^{\mathcal{C}' \cup \{i\}})^{-1}$ and, for $j \in \mathcal{C}'$, $\lambda_j^i := \lambda_j^{\mathcal{C}' \cup \{i\}}$ – both relative to the fixed index set $\mathcal{C}' \cup \{i\}$ – so that $\vec{s}_i = \mu_i \cdot \left(\vec{s} - \sum_{j \in \mathcal{C}'} \lambda_j^i \vec{s}_j \right)$. Consequently, the partial decryption of a ciphertext $\text{ct} = (c_0, \vec{c}_1)$ for an honest party $i \notin \mathcal{C}$ can be expressed as:

$$d_{\text{ct},i} = \vec{c}_1^\top \vec{s}_i + f_{\text{ct},i} = \mu_i \cdot \vec{c}_1^\top \vec{s} + f_{\text{ct},i} - \mu_i \cdot \vec{c}_1^\top \left(\sum_{j \in \mathcal{C}'} \lambda_j^i \vec{s}_j \right),$$

Furthermore, the first term involving the secret $\vec{s}$ can be expanded by substituting $\vec{b} = -A\vec{s} + \vec{e} \pmod{q}$ and $\vec{c}_1 = A^\top \vec{r}_2 + \vec{r}_1$:

$$\vec{c}_1^\top \vec{s} = (A^\top \vec{r}_2 + \vec{r}_1)^\top \vec{s} = \vec{r}_2^\top (-\vec{b} + \vec{e}) + \vec{r}_1^\top \vec{s} = -\vec{r}_2^\top \vec{b} + \vec{r}_2^\top \vec{e} + \vec{r}_1^\top \vec{s},$$

where $\vec{r}_2^\top \vec{b}$ is computable from the encryption randomness and the ciphertext.

Thus, we finally have the following expression for the partial decryption of an honest party $i \notin \mathcal{C}'$:

$$d_{\text{ct},i} = -\mu_i \vec{r}_2^\top \vec{b} + \mu_i (\vec{r}_2^\top \vec{e} + \vec{r}_1^\top \vec{s}) + f_{\text{ct},i} - \mu_i \cdot \vec{c}_1^\top \left(\sum_{j \in \mathcal{C}'} \lambda_j^i \vec{s}_j \right).$$

The change from Game_1 to Game_2 is purely syntactic, and thus $p_2 = p_1$.

Game₃: In this game, we modify the key generation algorithm by sampling the shares of the parties in $\mathcal{C}'$ uniformly at random in $\mathcal{R}_q^k$, instead of generating them via Shamir Secret Sharing, and we drop the remaining shares of the honest parties.

In Game_2 , since there are exactly $t-1$ parties in $\mathcal{C}'$, their shares are distributed uniformly at random in $\mathcal{R}_q^k$ and independent of the secret $\vec{s}$. Furthermore, the

shares of parties outside $\mathcal{C}'$ are no longer used in the game, and thus can be safely dropped. Therefore, the distribution of the adversary's view is unchanged in Game_3 compared to Game_2 :

$$p_3 = p_2.$$

Game₄: In this game, we replace the public key vector $\vec{b} = -A\vec{s} + \vec{e} \pmod{q}$ in Game_3 with a uniformly random value in $\mathcal{R}_q^k$ in Game_4 .

We wish to show that if the AHMLWE assumption holds, then Game_3 is computationally indistinguishable from Game_4 .

Observe that except for their use in $\vec{b}$, the secret vectors $\vec{s}$ and $\vec{e}$ were only used in the computation of the partial decryptions for honest parties, with the term $\mu_i(\vec{r}_2^\top \vec{e} + \vec{r}_1^\top \vec{s}) + f_{\text{ct},i}$ for $i \notin \mathcal{C}'$. Conveniently, this term can be seen as the response to a query to the hint oracle $\mathcal{O}_{\text{hint}}$ of the challenger for $\text{AHMLWE}_{q,k,k,Q,\sigma_s,\sigma_f,\mathcal{V}_B}$, with input $\vec{v} := \mu_i \cdot (\vec{r}_2, -\vec{r}_1) \in \mathcal{V}_B$, where B is to be defined. We then set $\vec{b}$ to the MLWE sample in the AHMLWE instance, identifying $\vec{e}$ with the first half and $-\vec{s}$ with the second half of the AHMLWE secret, since $\vec{b} = -A\vec{s} + \vec{e}$.

Let us now define B . Recall $\|\vec{r}\|_2 \leq e^{1/4}\sigma_r\sqrt{N(2k+1)}$: with certainty when ct is produced by EncO (the adversary is required to supply $\vec{r} \in \mathcal{RD}$), and with overwhelming probability when ct is produced by ChalO ($\vec{r}$ is honestly sampled from $D_{\sigma_r}^{2k+1}$, by Lem. 3 directly). Since $(\vec{r}_2, \vec{r}_1)$ is a subset of the coordinates of $\vec{r}$, $\|(\vec{r}_2, \vec{r}_1)\|_2 \leq \|\vec{r}\|_2$, so by Corollary 1's Cauchy-Schwarz step, $\sum_i \|r_i\|_1^2 \leq N\|(\vec{r}_2, \vec{r}_1)\|_2^2 \leq e^{1/2}N^2(2k+1)\sigma_r^2$ for the $2k$ coordinates r_i of $(\vec{r}_2, \vec{r}_1)$.

Together with $\|\mu_i\|_1 \leq B_{\lambda^{-1}}$ (Lem. 5) and submultiplicativity of the one-norm, this gives $\sum_u \|v_u\|_1^2 \leq B_{\lambda^{-1}}^2 \cdot e^{1/2}N^2(2k+1)\sigma_r^2$ for the $2k$ coordinates v_u of $\vec{v}$, so that $\vec{v} \in \mathcal{V}_B$ for $B = e^{1/2}N^2(2k+1)(B_{\lambda^{-1}}\sigma_r)^2$.

Note that we crucially rely on the adaptive nature of the AHMLWE problem, as the vector $\vec{v}$ is determined by the encryption randomness $\vec{r}$ of the ciphertext, which is chosen adaptively by the adversary $\mathcal{A}$ when it queries EncO – in that case, $\vec{r}$ need not follow the prescribed distribution D_{σ_r} , only the bound enforced by $\mathcal{RD}$, which is exactly what the bound on $\vec{v}$ above relies on instead.

Hence, taking a union bound over the (at most Q_c) challenge ciphertexts for which $\|\vec{r}\|_2 \leq e^{1/4}\sigma_r\sqrt{N(2k+1)}$ may fail to hold,

$$|p_3 - p_4| \leq \text{Adv}_{q,k,k,Q,\sigma_s,\sigma_f,\mathcal{V}_B}^{\text{AHMLWE}}(\mathcal{B}) + Q_c \cdot 2^{-N(2k+1)/10},$$

where $\mathcal{B}$ is the adversary for the AHMLWE problem that internally simulates $\mathcal{A}$ and answers its queries as described above.

Remark 1. The vectors $\vec{s}$ and $\vec{e}$ will continue to be sampled by the games, even though they are no longer related to the public key. They will remain important for the simulation of the partial decryptions.

Game₅: In this game, we equivalently rewrite the partial decryptions of a challenge ciphertext $\text{ct} = (c_0, \vec{c}_1)$ as a function of the challenge ciphertexts, shares of parties in $\mathcal{C}'$, together with a small hint vector on the ciphertext randomness.

First, the partial decryption of a party $i \in \mathcal{C}'$ remains $d_{\text{ct},i} = \vec{c}_1^\top \vec{s}_i + f_{\text{ct},i}$, which is already of the desired form.

For a party $i \notin \mathcal{C}'$, we use the fact that $\vec{r}_2^\top \vec{b} = c_0 - r_0 - \gamma^{-1} \cdot \lfloor \frac{q}{p} \rfloor m$ and $\vec{c}_1 = A^\top \vec{r}_2 + \vec{r}_1 \pmod{q}$ to rewrite the partial decryption as

$$\begin{aligned} d_{\text{ct},i} &= \mu_i(-\vec{r}_2^\top \vec{b} + \vec{r}_2^\top \vec{e} + \vec{r}_1^\top \vec{s}) + f_{\text{ct},i} - \mu_i \cdot \vec{c}_1^\top \left(\sum_{j \in \mathcal{C}'} \lambda_j^i \vec{s}_j \right) \\ &= -\mu_i(c_0 - \gamma^{-1} \cdot \lfloor \frac{q}{p} \rfloor m) + \underbrace{\mu_i(r_0 + \vec{r}_2^\top \vec{e} + \vec{r}_1^\top \vec{s}) + f_{\text{ct},i}}_{\text{hint on } (\vec{r}_0, \vec{r}_2, \vec{r}_1)} - \mu_i \cdot \vec{c}_1^\top \left(\sum_{j \in \mathcal{C}'} \lambda_j^i \vec{s}_j \right). \end{aligned}$$

This is again a purely syntactic change, and thus $p_5 = p_4$.

Game₆: In this game, we replace the challenge ciphertexts produced by ChalO with uniformly random values in $\mathcal{R}_q^{k+1}$. Note that, as for vectors $\vec{s}, \vec{e}$, the game keeps sampling a corresponding randomness $\vec{r} = (\vec{r}_0, \vec{r}_1, \vec{r}_2)$ for the challenge ciphertexts, as it is needed for the simulation of the partial decryptions. However, the ciphertexts produced by ChalO themselves are now independent of $\vec{r}$.

We argue that this change is indistinguishable under the AHMLWE assumption. Indeed, we can see a challenge ciphertext ct as an MLWE sample with secret $\vec{r}_2$ and error $(r_0, \vec{r}_1)$ for the public matrix $A' = (b, A^\top)$, where b is uniformly random in $\mathcal{R}_q^k$. Furthermore, except for the computation of ct , the randomness $\vec{r}$ of the challenge ciphertexts is only used in the simulation of the partial decryptions for parties $j \notin \mathcal{C}'$, with the term $\mu_i(r_0 + \vec{r}_2^\top \vec{e} + \vec{r}_1^\top \vec{s}) + f_{\text{ct},i}$ for $i \notin \mathcal{C}'$. Remark that for a given ct , at most n such terms are ever needed, since outputs of $\text{PartDecO}(\text{ct}, i)$ are memoized. Hence this term can be seen as the response to a query to the hint oracle $\mathcal{O}_{\text{hint}}$ of the challenger for $\text{AHMLWE}_{q,k+1,k,n,\sigma_r,\sigma_f,\mathcal{V}_B}$, with input $\vec{v} := \mu_i \cdot (1, \vec{s}, \vec{e})$.

Unlike Game₄, the $k+k$ coordinates of $\vec{v}$ built from $(\vec{s}, \vec{e})$ are not adversarially chosen: $\vec{s}, \vec{e} \leftarrow D_{\sigma_s}^k$ are honestly sampled and independent of μ_i . By Corollary 1 applied to $(\vec{s}, \vec{e}) \leftarrow D_{\sigma_s}^{2k}$, except with probability at most $2^{-2kN/10}$, $\sum_u \|s_u\|_1^2 \leq e^{1/2} \cdot 2kN^2 \sigma_s^2$ for the $2k$ coordinates s_u of $(\vec{s}, \vec{e})$; by one-norm submultiplicativity, the corresponding coordinates of $\vec{v}$ then satisfy

$$\sum_u \|v_u\|_1^2 \leq \|\mu_i\|_1^2 \cdot \sum_u \|s_u\|_1^2 \leq e^{1/2} \cdot 2kN^2 \cdot B_{\lambda^{-1}}^2 \sigma_s^2.$$

Together with the first coordinate of $\vec{v}$ (the constant μ_i itself, with one-norm squared at most $B_{\lambda^{-1}}^2$, by Lem. 5), this shows that, except with probability at most $2^{-2kN/10}$ over the sampling of $\vec{s}, \vec{e}$, $\vec{v} \in \mathcal{V}_B$ for

$$B = B_{\lambda^{-1}}^2 \left(1 + 2e^{1/2} kN^2 \sigma_s^2 \right).$$

If this (negligible-probability) event fails, we bound the adversary's advantage trivially by 1.

Applying this argument to all challenge ciphertexts, we conclude that if the AHMLWE assumption holds, then Game₅ is computationally indistinguishable

from Game_6 :

$$|p_5 - p_6| \leq Q_c \cdot \text{Adv}_{q,k+1,k,n,\sigma_r,\sigma_f,\mathcal{V}_B}^{\text{AHMLWE}}(\mathcal{B}_2) + 2^{-2kN/10},$$

where $\mathcal{B}_2$ is the adversary for the AHMLWE problem that internally simulates $\mathcal{A}$ and answers its queries as described above.

Game₇: This hybrid performs syntactical rewrites to prepare for the application of the AHMLWE assumption in the Game_9 hybrid to argue that the partial decryptions of honest parties in $\mathcal{C}' \setminus \mathcal{C}$ for challenge ciphertexts are computationally independent of the challenge ciphertexts themselves.

Recall that the partial decryption of a challenge ciphertext ct for a party $i \in \mathcal{C}' \setminus \mathcal{C}$ is $d_{\text{ct},i} = \vec{c}_1^\top \vec{s}_i + f_{\text{ct},i}$, for the uniformly sampled $\vec{c}_1$. In order to later interpret this as an AHMLWE sample (using $\vec{c}_1$ as public matrix, and $\vec{s}_i$ as secret), we need to gather all dependencies on the secret shares $\vec{s}_i$ of parties in $\mathcal{C}' \setminus \mathcal{C}$, and either ensure they can be interpreted either as a MLWE sample, or as a hint. These dependencies are all within the partial decryption oracle PartDecO .

We shall rely on the following samples derived from $\vec{s}_i$, for $i \in \mathcal{C}'$:

- We introduce samples $\vec{b}_i = A \cdot \vec{s}_i + \gamma \vec{e}_i$ for $i \in \mathcal{C}'$, where $\vec{e}_i \leftarrow \mathcal{U}(\mathcal{R}_q^k)$ is a fresh uniformly random error term. Note that the use of a uniform noise will be explicitly handled in Game_9 when invoking AHMLWE.
- We will also rely on the partial decryptions of challenge ciphertexts ct produced by ChalO , for parties $i \in \mathcal{C}'$: $d_{\text{ct},i} = \vec{c}_1^\top \vec{s}_i + f_{\text{ct},i}$.

We distinguish the case where PartDecO is queried for a ciphertext generated by ChalO or EncO , and whether $i \in \mathcal{C}'$ or $i \notin \mathcal{C}'$. In Cases 2 and 4 below, we repeatedly use the linearity $\vec{c}_1^\top (\sum_{j \in \mathcal{C}'} \lambda_j^i \vec{s}_j) = \sum_{j \in \mathcal{C}'} \lambda_j^i \vec{c}_1^\top \vec{s}_j$, substituting $\vec{c}_1^\top \vec{s}_j$ either by the MLWE sample of Case 1 or by $d_{\text{ct},j} - f_{\text{ct},j}$.

Case 1, ct generated by EncO and $i \in \mathcal{C}'$: In this case, the partial decryption is $d_{\text{ct},i} = \vec{c}_1^\top \vec{s}_i + f_{\text{ct},i}$, with $\vec{c}_1 = A^\top \vec{r}_2 + \vec{r}_1$ for some $\vec{r} \in \mathcal{RD}$. We can rewrite this as

$$\begin{aligned} d_{\text{ct},i} &= \vec{c}_1^\top \vec{s}_i + f_{\text{ct},i} \\ &= (A^\top \vec{r}_2 + \vec{r}_1)^\top \vec{s}_i + f_{\text{ct},i} \\ &= \vec{r}_2^\top \underbrace{(A \cdot \vec{s}_i + \gamma \vec{e}_i)}_{\vec{b}_i} + (\vec{r}_1^\top \vec{s}_i - \vec{r}_2^\top \gamma \vec{e}_i + f_{\text{ct},i}), \end{aligned}$$

Case 2, ct generated by EncO and $i \notin \mathcal{C}'$:

Recall that in Game_2 ,

$$\begin{aligned} d_{\text{ct},i} &= -\mu_i \vec{r}_2^\top \vec{b} + \mu_i (\vec{r}_2^\top \vec{e} + \vec{r}_1^\top \vec{s}) + f_{\text{ct},i} \\ &\quad - \mu_i \cdot \vec{c}_1^\top \left(\sum_{j \in \mathcal{C}'} \lambda_j^i \vec{s}_j \right) \end{aligned}$$

Substituting $\vec{c}_1^\top \vec{s}_j = \vec{r}_2^\top \vec{b}_j + (\vec{r}_1^\top \vec{s}_j - \vec{r}_2^\top \gamma \vec{e}_j)$ from Case 1,

$$\vec{c}_1^\top \left(\sum_{j \in \mathcal{C}'} \lambda_j^i \vec{s}_j \right) = \vec{r}_2^\top \sum_{j \in \mathcal{C}'} \lambda_j^i \vec{b}_j + \sum_{j \in \mathcal{C}'} \lambda_j^i (\vec{r}_1^\top \vec{s}_j - \vec{r}_2^\top \gamma \vec{e}_j).$$

Hence, the partial decryption can be expressed as a function of $\vec{b}$, $\vec{r}$, $\vec{e}$, $\vec{s}$, the MLWE samples $\vec{b}_i = A \cdot \vec{s}_i + \gamma \vec{e}_i$ for $i \in \mathcal{C}'$, to which we add a hint vector on the ciphertext randomness:

$$-\mu_i \cdot \sum_{j \in \mathcal{C}'} \lambda_j^i (\vec{r}_1^\top \vec{s}_j - \vec{r}_2^\top \gamma \vec{e}_j) + f_{\text{ct},i}.$$

Case 3, ct generated by ChalO and $i \in \mathcal{C}'$: In this case, the partial decryption is $d_{\text{ct},i} = \vec{c}_1^\top \vec{s}_i + f_{\text{ct},i}$. This is directly of the desired form, as **ct** is a challenge ciphertext.

Case 4, ct generated by ChalO and $i \notin \mathcal{C}'$: In this case, due to $\mathfrak{D}_5$, the partial decryption is

$$\begin{aligned} d_{\text{ct},i} &= -\mu_i (c_0 - \gamma^{-1} \cdot \lfloor \frac{q}{p} \rfloor m) + \mu_i (r_0 + \vec{r}_2^\top \vec{e} + \vec{r}_1^\top \vec{s}) + f_{\text{ct},i} \\ &\quad - \mu_i \cdot \vec{c}_1^\top \left(\sum_{j \in \mathcal{C}'} \lambda_j^i \vec{s}_j \right) \end{aligned}$$

Substituting $\vec{c}_1^\top \vec{s}_j = d_{\text{ct},j} - f_{\text{ct},j}$,

$$\vec{c}_1^\top \left(\sum_{j \in \mathcal{C}'} \lambda_j^i \vec{s}_j \right) = \sum_{j \in \mathcal{C}'} \lambda_j^i (d_{\text{ct},j} - f_{\text{ct},j}).$$

Hence, the partial decryption can be expressed as a function of $\vec{b}$, $\vec{r}$, $\vec{e}$, $\vec{s}$, c_0 , m , the partial decryptions $d_{\text{ct},j}$ of the challenge ciphertext for parties $j \in \mathcal{C}'$, to which we add:

$$\mu_i \cdot \sum_{j \in \mathcal{C}'} \lambda_j^i f_{\text{ct},j} + f_{\text{ct},i}.$$

This change is purely syntactic, and thus $p_7 = p_6$.

Game₈: This game is also just a preparation to apply AHMLWE in the next hybrid.

In this game, for each $i \in \mathcal{C}' \setminus \mathcal{C}$, we rewrite the share $\vec{s}_i$ and the error terms $(f_{\text{ct},i})_{\text{ct}}$ (**ct** produced by **ChalO**) introduced in **Game₇**, pulling out a factor of γ from each so that the coefficient $\mu_i(\gamma \lambda_j^i)$ multiplying them in the partial decryption simulation (see **Game₉**) admits a small representative.

First, since $\vec{s}_i$ is sampled uniformly in $\mathcal{R}_q^k$ and multiplication by the invertible γ is a bijection of $\mathcal{R}_q$, we may equivalently sample a fresh $\vec{s}'_i \leftarrow \mathcal{U}(\mathcal{R}_q^k)$ and set $\vec{s}_i := \gamma \vec{s}'_i$.

Second, provided $\sigma_f \geq \sqrt{2/\pi} \cdot \gamma \eta_\varepsilon(\mathbb{Z}^N)$, set

$$\sigma_1^2 := \frac{\sigma_f^2}{2\gamma^2} \left(1 + \sqrt{1 - \frac{2\gamma^2 \eta_\varepsilon(\mathbb{Z}^N)^2}{\pi \sigma_f^2}} \right), \quad \sigma_2^2 := \frac{\sigma_f^2}{2} \left(1 - \sqrt{1 - \frac{2\gamma^2 \eta_\varepsilon(\mathbb{Z}^N)^2}{\pi \sigma_f^2}} \right),$$

so that $1/\sigma_1^2 + \gamma^2/\sigma_2^2 = 2\pi/\eta_\varepsilon(\mathbb{Z}^N)^2$ and $\gamma^2 \sigma_1^2 + \sigma_2^2 = \sigma_f^2$ – this is the largest value of σ_1 for which such a σ_2 exists. By **Lem. 6** applied with $\mathbf{M} = \gamma \mathbf{I}_N$, the

distribution of $\gamma g_1 + g_2$, for independent $g_1 \leftarrow D_{\sigma_1}$ and $g_2 \leftarrow D_{\sigma_2}$, is within statistical distance 4ε of D_{σ_f} .

We replace each coefficient of each $f_{\text{ct},i}$, with an independent fresh sample of this form: writing $f_{\text{ct},i} := \gamma f'_{\text{ct},i} + f''_{\text{ct},i}$ for $f'_{\text{ct},i} \leftarrow D_{\sigma_1}$ and $f''_{\text{ct},i} \leftarrow D_{\sigma_2}$, independently for every $i \in \mathcal{C}' \setminus \mathcal{C}$ and every ciphertext. By a union bound over the at most $(t-1)Q_c$ coefficients so replaced,

$$|p_7 - p_8| \leq 4(t-1)Q_c\varepsilon.$$

Game₉: In this game, we replace the MLWE samples $\vec{b}_i$ and $d_{\text{ct},i}$ defined in **Game₇** with uniformly random values in $\mathcal{R}_q^k$ and $\mathcal{R}_q$, respectively, for $i \in \mathcal{C}' \setminus \mathcal{C}$ and challenge ciphertexts ct produced by **ChalO**.

We can argue that this change is indistinguishable under the AHMLWE assumption. We proceed by replacing these samples iteratively for each $i \in \mathcal{C}' \setminus \mathcal{C}$. Indeed, writing $\vec{b}_i = \gamma(A\vec{s}'_i + \vec{e}_i)$ and $d_{\text{ct},i} = \gamma(\vec{c}_1^\top \vec{s}'_i + f'_{\text{ct},i}) + f''_{\text{ct},i}$, we can see the vector $(\vec{b}_i, (d_{\text{ct},i})_{\text{ct}})$ produced by **ChalO** as γ times an MLWE sample with secret $\vec{s}'_i$ and error $(\vec{e}_i, (f'_{\text{ct},i})_{\text{ct}})$ for the public matrix $A' = (A, (\vec{c}_1)_{\text{ct}})$, shifted by the independent, freshly-sampled $(f''_{\text{ct},i})_{\text{ct}}$. Since we only consider shares of parties $i \notin \mathcal{C}$, the shares $\vec{s}'_i$ and $\vec{e}_i$ are not directly revealed to the adversary, and except for the aforementioned samples, they (together with $(f'_{\text{ct},i})_{\text{ct}}$) are only used in the simulation of the partial decryptions:

- $\gamma(\vec{r}_1^\top \vec{s}'_i - \vec{r}_2^\top \vec{e}_i) + f_{\text{ct},i}$ for partial decryptions of ciphertexts generated by **EncO** and parties $i \in \mathcal{C}'$.
- $-\mu_i(\gamma\lambda_j^i)(\vec{r}_1^\top \vec{s}'_j - \vec{r}_2^\top \vec{e}_j) + f_{\text{ct},i}$ for partial decryptions of ciphertexts generated by **EncO** and parties $i \notin \mathcal{C}'$.
- $\mu_i(\gamma\lambda_j^i)f'_{\text{ct},j} + \mu_i\lambda_j^i f''_{\text{ct},j} + f_{\text{ct},i}$ for partial decryptions of ciphertexts generated by **ChalO** and parties $i \notin \mathcal{C}'$.

The terms involving $f''_{\text{ct},j}$ are publicly computable (as discussed above), so we fold them directly into the simulator's response without appealing to any hardness assumption.

This is almost directly an instance of the AHMLWE problem, except that we defined this problem with a single distribution for all secret polynomials, while here $\vec{s}'_i, \vec{e}_i$ are uniform and the errors $f'_{\text{ct},i}$ are D_{σ_1} -distributed. However, since $\vec{s}'_i, \vec{e}_i$ are sampled uniformly over $\mathcal{R}_q^k$, it is easy to see that distinguishing the above samples from uniform is at least as hard as distinguishing in the case where all the secret polynomials are sampled from the same distribution D_{σ_1} . This is because $\mathcal{U}(\mathcal{R}_q) = \mathcal{U}(\mathcal{R}_q) + D_{\sigma_1} \bmod q$ statically, so any adversary breaking the uniform-secret game can be used to break the D_{σ_1} -secret AHMLWE game: given a Gaussian-secret MLWE sample, add a fresh uniform shift to the secret to obtain a uniform-secret sample, then run the adversary. Hence the uniform-secret game is no easier than AHMLWE.

Each of the three bullets above (restricted to its hardness-relevant part) is itself a hint query on the secret, which has two parts: $(\vec{s}'_i, \vec{e}_i)$, and the Q_c coordinates $(f'_{\text{ct},i})_{\text{ct}}$, one for each challenge ciphertext. The first two bullets query

the $(\vec{s}_i, \vec{e}_i)$ part, with coefficient vector $\gamma(\vec{r}_1, -\vec{r}_2)$ for the first and $\mu_i(\gamma\lambda_j^i) \cdot (\vec{r}_1, -\vec{r}_2)$ for the second; the third instead queries a single coordinate $f'_{\text{ct},j}$ of the second part, with the scalar coefficient $\mu_i(\gamma\lambda_j^i)$.

The first two bullets only arise for ciphertexts produced by EncO , where, as in Game_4 , $\vec{r}_1, \vec{r}_2$ are supplied by the adversary from the valid randomness space $\mathcal{RD}$, giving (deterministically, as $(\vec{r}_1, \vec{r}_2)$ ranges over a subset of $\vec{r}$'s coordinates, via the same Cauchy-Schwarz step as in Game_4) the bound $\sum_u \|r_u\|_1^2 \leq e^{1/2} N^2 (2k+1) \sigma_r^2$ for the $2k$ coordinates r_u of $(\vec{r}_1, \vec{r}_2)$.

The first bullet's coefficient $\gamma(\vec{r}_1, -\vec{r}_2)$ therefore satisfies $\sum_u \|v_u\|_1^2 \leq \gamma^2 e^{1/2} N^2 (2k+1) \sigma_r^2$ by submultiplicativity. Together with $\|\mu_i(\gamma\lambda_j^i)\|_1 \leq B_{\lambda^{-1}} B_\lambda$ by submultiplicativity, where $B_\lambda, B_{\lambda^{-1}}$ are as in Lem. 5, the second bullet's query satisfies $\sum_u \|v_u\|_1^2 \leq e^{1/2} N^2 (2k+1) (B_{\lambda^{-1}} B_\lambda \sigma_r)^2$. The third bullet's query, the single coordinate $\mu_i(\gamma\lambda_j^i)$ itself, satisfies $\|v\|_1^2 \leq (B_{\lambda^{-1}} B_\lambda)^2$, which is dominated by the second bullet's bound (as $e^{1/2} N^2 (2k+1) \sigma_r^2 \geq 1$ for any meaningful choice of parameters).

So we may take $\mathcal{V}_B$ for $B = e^{1/2} N^2 (2k+1) \sigma_r^2 \max(\gamma, B_{\lambda^{-1}} B_\lambda)^2$ as a valid challenge set covering every hint query that arises in this game, on either part of the secret.

Formally, there exists an adversary $\mathcal{B}_3$ for the $\text{AHMLWE}_{q,k+Q_c,k,Q,\sigma_1,\sigma_f,\mathcal{V}_B}$ problem with up to Q hint queries that internally simulates $\mathcal{A}$ and answers its queries as described above, such that

$$|p_8 - p_9| \leq (t-1) \cdot \text{Adv}_{q,k+Q_c,k,Q,\sigma_1,\sigma_f,\mathcal{V}_B}^{\text{AHMLWE}}(\mathcal{B}_3).$$

Game₁₀: In this game, we focus on the partial decryptions of the challenge ciphertexts. For the first $t-1-|\mathcal{C}|$ queries for parties $j \notin \mathcal{C}$, we return a uniformly random value in $\mathcal{R}_q$ instead of the simulated partial decryption. Then, we shall answer the remaining queries to preserve consistency with the previous game.

Recall that, by construction, Game_9 produces the partial decryptions of a challenge ciphertext $\text{ct} = (c_0, \vec{c}_1)$ for *every* party $i \in [n]$ as follows:

- $d_{\text{ct},i} = \vec{c}_1^\top \vec{s}_i + f_{\text{ct},i}$ for $i \in \mathcal{C}$ – the real partial decryption, computable by the adversary itself from its corrupted share $\vec{s}_i$.
- $d_{\text{ct},i} \leftarrow \mathcal{R}_q$ for $i \in \mathcal{C}' \setminus \mathcal{C}$.
- $d_{\text{ct},i} = -\mu_i(c_0 - \gamma^{-1} \cdot \lfloor \frac{q}{p} \rfloor m) + \mu_i(r_0 + \vec{r}_2^\top \vec{e} + \vec{r}_1^\top \vec{s}) + f_{\text{ct},i} - \mu_i \cdot \sum_{j \in \mathcal{C}'} \lambda_j^i (d_{\text{ct},j} - f_{\text{ct},j})$ for $i \notin \mathcal{C}'$.

We notice that $d_{\text{ct},i} - f_{\text{ct},i}$ can be expressed as evaluations of a single degree- $< t$ polynomial Q_{ct} such that:

- $Q_{\text{ct}}(0) = c_0 - \gamma^{-1} \cdot \lfloor \frac{q}{p} \rfloor m + r_0 + \vec{r}_2^\top \vec{e} + \vec{r}_1^\top \vec{s}$.
- $Q_{\text{ct}}(\alpha_i) = \vec{c}_1^\top \vec{s}_i$ for $i \in \mathcal{C}$.
- $Q_{\text{ct}}(\alpha_i) = d_{\text{ct},i} - f_{\text{ct},i}$ for $i \in \mathcal{C}' \setminus \mathcal{C}$.

Indeed, for $i \notin \mathcal{C}'$, since $Q_{\text{ct}}(0) = \sum_{j \in \mathcal{C}'} \lambda_j^i Q_{\text{ct}}(\alpha_j) + \mu_i^{-1} Q_{\text{ct}}(\alpha_i)$, we can solve for $Q_{\text{ct}}(\alpha_i)$ and obtain exactly the formula for $d_{\text{ct},i} - f_{\text{ct},i}$ above.

Only the first two bullets are actual constraints: $Q_{\text{ct}}(0)$ and $\{Q_{\text{ct}}(\alpha_i)\}_{i \in \mathcal{C}}$ are fixed by the vectors $\vec{e}, \vec{s}, \vec{r}$, the message, and corrupted shares, while the third bullet samples Q_{ct} uniformly at the remaining $t - 1 - |\mathcal{C}|$ positions $\mathcal{C}' \setminus \mathcal{C}$. But Q_{ct} has degree $< t$, so once its value at 0 and $\mathcal{C}$ is fixed, it has exactly $t - 1 - |\mathcal{C}|$ degrees of freedom left, regardless of where we choose to read them off.

Since $\mathcal{S}$ gives valid Lagrange coefficients for every size- t index set, evaluating such a polynomial at *any* $t - 1 - |\mathcal{C}|$ further points yields values that are uniform in $\mathcal{R}_q$ and mutually independent, whichever points are chosen. So the role played by $\mathcal{C}' \setminus \mathcal{C}$ in the second bullet can equally well be played by any other $t - 1 - |\mathcal{C}|$ honest parties – in particular, by the first $t - 1 - |\mathcal{C}|$ honest parties the adversary happens to query, exactly as Game_{10} does. This only changes which evaluations of Q_{ct} are sampled directly and which are recovered by interpolation, not the resulting distribution of Q_{ct} , and hence not that of the partial decryptions $d_{\text{ct},i} = Q_{\text{ct}}(\alpha_i) + f_{\text{ct},i}$. Therefore:

$$p_{10} = p_9.$$

The simulator and conclusion. By Game_{10} , every oracle response can be produced without ever using an honest party's secret key share, and the only point where the message m of a challenge ciphertext is needed is to answer a partial decryption query once the threshold has effectively been met. This is exactly the requirements on the simulator $\mathcal{S}$ of SemiMalSimCPA^1 .

We now define $\mathcal{S}$ to match the adversary's view in Game_{10} : on input $(1^\lambda, t, n, \mathcal{C})$, $\mathcal{S}$ samples $\text{pk} = (\vec{b}, A)$ uniformly at random (c.f. Game_4) and $(\vec{s}_k, \alpha_k)_{k \in \mathcal{C}}$ uniformly at random (c.f. Game_3), and additionally samples $\vec{s}, \vec{e} \leftarrow D_{\sigma_s}^k$ for its own internal use only.

On an $\text{EncO}(m, \text{rnd})$ query, $\mathcal{S}$ simply records $(m, \vec{r})$; on a ChalO query, $\mathcal{S}$ samples a uniformly random ciphertext $\text{ct} = (c_0, \vec{c}_1)$ (c.f. Game_6), together with bookkeeping randomness $\vec{r} = (r_0, \vec{r}_1, \vec{r}_2)$, and records $(\text{ct}, \vec{r})$. On $\text{PartDecO}(\text{ct}, j)$ with $j \notin \mathcal{C}$, $\mathcal{S}$ answers as follows:

- if $\text{ct} \in L_{\text{Enc}}$, using the AHMLWE-hint-style formula of $\text{Game}_5/\text{Game}_7$ applied to the recorded $\vec{r}$ (and to $\vec{s}, \vec{e}$, and to the corrupted shares $\vec{s}_i, \vec{e}_i, i \in \mathcal{C}'$);
- if $\text{ct} \in L_{\text{Chal}}$, we sample $f_{\text{ct},i}$ once and reuse it on any later (ct, i) query, and answer as follows:
 - if $i \in \mathcal{C}$, we return $\vec{c}_1^\top \vec{s}_i + f_{\text{ct},i}$;
 - if $i \notin \mathcal{C}$ but is among the first $t - 1 - |\mathcal{C}|$ honest parties queried on ct , we return a uniform value in $\mathcal{R}_q$, sampled once and reused on any later (ct, i) query (c.f. $\text{Game}_9/\text{Game}_{10}$);
 - otherwise (i.e., once $|P[\text{ct}] \cup \mathcal{C}| \geq t$), using $m = M[\text{ct}]$ together with the now-fixed values $\{d_{\text{ct},j}\}_{j \in \mathcal{C}'}$ to compute the forced interpolated value $d_{\text{ct},i}$ of Game_{10} .

The last case matches exactly when SemiMalSimCPA^1 itself hands $\mathcal{S}$ the message $M[\text{ct}]$.

By construction, the view of $\mathcal{A}$ when run against this $\mathcal{S}$ in SemiMalSimCPA^1 is identically distributed to the view of $\mathcal{A}$ in Game_{10} , so $\Pr[\text{SemiMalSimCPA}_{\text{tPKE}, \mathcal{A}, \mathcal{S}}^1(1^\lambda) =$

$1] = p_{10}$. Together with $p_0 = \Pr[\text{SemiMalSimCPA}_{\text{tPKE}, \mathcal{A}, \mathcal{S}}^0(1^\lambda) = 1]$ (i.e., $|p_0 - p_{10}| = \text{Adv}_{\text{tPKE}}^{\text{SemiMalSimCPA}}(\mathcal{A}, \mathcal{S})$ by definition) and the triangle inequality over $|p_0 - p_1|, \dots, |p_9 - p_{10}|$ —all of which are either 0 or bounded above in the corresponding game, we recover the theorem statement. $\square$

5 Achieving CCA Security

In this section, we extend our **SemiMalSimCPA**-secure tPKE to achieve **SimCCA** security. We rely on the BKW transform [BKW26], which generically upgrades a semi-malicious secure tPKE to a CCA-secure one by leveraging a simulation-extractable Non-Interactive Zero-Knowledge (NIZK) proof. Roughly, this transform appends a NIZK proof of well-formedness to each ciphertext, which is verified before partial decryption. Simulation extractability ensures that we can:

- Simulate the proofs attached to challenge ciphertexts so that they do not leak any secret information.
- Extract the randomness and message underlying all other ciphertexts, which is necessary to simulate their partial decryptions in the **SemiMalSimCPA** model.

We discuss how to efficiently instantiate a simulation-extractable NIZK from lattice assumptions and finally provide a concrete CCA-secure tPKE construction.

5.1 Non-interactive Proof System

We first recall the notion of *non-interactive zero-knowledge proof* (NIZK) systems in Def. 10.

Definition 10 (NIZK). *A non-interactive zero-knowledge proof system for a language $\mathcal{L}$ in the common reference string (CRS) model is a tuple $\text{NIZK} = (\text{Setup}, \text{Prove}, \text{Verify})$ with the following syntax:*

- $\text{Setup}(1^\lambda) \rightarrow \text{crs}$ *Generates a common reference string.*
- $\text{Prove}(\text{crs}, x, w) \rightarrow \pi$ *Given a CRS crs, and $(x, w) \in \mathcal{L}$, outputs a proof π .*
- $\text{Verify}(\text{crs}, x, \pi) \rightarrow b$ *Given a CRS crs, a statement x , and a proof π , outputs a verdict $b \in \{0, 1\}$.*

We require NIZK to be perfectly correct for $\mathcal{L}$, meaning that for all $(x, w) \in \mathcal{L}$, and all $\text{crs} \leftarrow \text{Setup}(1^\lambda)$, $\text{Verify}(\text{crs}, x, \pi) = 1$ where $\pi \leftarrow \text{Prove}(\text{crs}, x, w)$.

We now introduce the property of *simulation extractability* for a NIZK, required by the CPA-to-CCA tPKE transform. Informally, a simulation-extractable NIZK is equipped with a simulator that produces convincing proofs without knowing any witness—so that proofs attached to challenge ciphertexts leak no secret information—while an extractor is still able to recover a valid witness from any proof that the simulator did not itself produce. This single notion thus captures both the (zero-knowledge) simulation-soundness and the straight-line

extractability that the transform relies on. Note that there exists a weak variant of simulation extractability, which in particular tolerates proof malleability. However, since it is insufficient for the BKW transform, we do not recall it here.

Definition 11 (Simulation Extractability). A NIZK scheme $\text{NIZK} = (\text{Setup}, \text{Prove}, \text{Verify})$ for a language $\mathcal{L}$ with associated relation $\mathcal{R}$ is simulation extractable if there exists a PPT simulator $\text{Sim} = (\text{Sim.Setup}, \text{Sim.Sim}, \text{Sim.Ext})$ such that

$$|\Pr[\text{SIMEXT}_{\text{NIZK}, \text{Sim}, \mathcal{A}}^0(1^\lambda) = 1] - \Pr[\text{SIMEXT}_{\text{NIZK}, \text{Sim}, \mathcal{A}}^1(1^\lambda) = 1]|$$

is negligible for any PPT adversary $\mathcal{A}$, where the experiment $\text{SIMEXT}_{\text{NIZK}, \text{Sim}, \mathcal{A}}^b$ is defined in Fig. 4.

$\text{SIMEXT}_{\text{NIZK}, \text{Sim}, \mathcal{A}}^b(1^\lambda)$	$\text{ProveO}(x, w)$
1: if $b = 0$ then $\text{crs} \xleftarrow{\$} \text{NIZK.Setup}(1^\lambda)$	1: assert $(x, w) \in \mathcal{R}$
2: if $b = 1$ then $(\text{crs}, \text{st}) \xleftarrow{\$} \text{Sim.Setup}(1^\lambda)$	2: if $b = 0$ then $\pi \xleftarrow{\$} \text{Prove}(\text{crs}, x, w)$
3: $\mathcal{W} \leftarrow \emptyset$	3: if $b = 1$ then
4: return $\mathcal{A}^{\text{ProveO}, \text{VerifyO}}(\text{crs})$	4: $(\pi, \text{st}) \xleftarrow{\$} \text{Sim.Sim}(x, \text{st})$
	5: $\mathcal{W} \leftarrow \mathcal{W} \cup \{(x, \pi)\}$
	6: return π
	$\text{VerifyO}(x, \pi)$
	1: if $\text{Verify}(\text{crs}, x, \pi) = 0$ then return 0
	2: if $b = 0$ then return 1
	3: if $(x, \pi) \in \mathcal{W}$ then return 1
	4: $(w, \text{st}) \xleftarrow{\$} \text{Sim.Ext}(x, \pi, \text{st})$
	5: assert $(x, w) \in \mathcal{R}$ // no extraction failure
	6: return 1

Fig. 4. Experiment for the (strong) simulation extractability of a NIZK system.

5.2 Threshold CPA-to-CCA Transform

The Naor-Yung paradigm [NY90] transforms a CPA-secure PKE scheme into a CCA-secure one with the help of a simulation-sound NIZK [DNR04]. In its original form, a ciphertext consists of two encryptions $\mathbf{c}_1$ and $\mathbf{c}_2$ of the same message under two independent key pairs, together with a NIZK proof that they indeed encrypt the same plaintext; decryption verifies the proof and then decrypts either of the two components. This paradigm has been lifted to the threshold setting under additional conditions [FP01, BKW26]. We rely on the variant of Brzuska et al. [BKW26], which upgrades a SemiMalSimCPA -secure tPKE into a SimCCA -secure one by attaching to each ciphertext a *simulation-extractable* NIZK proof

of well-formedness, without resorting to double encryption. The transform is as follows.

Let $\text{tPKE}_0 = (\text{KeyGen}, \text{Enc}, \text{PartDec}, \text{FinDec})$ be a SemiMalSimCPA -secure tPKE with message space $\mathcal{M}$, public key pk_0 , and encryption randomness space $\mathcal{RD}$. The transform attaches to each ciphertext a proof that it is well-formed, i.e., that it lies in the image of $\text{tPKE}_0.\text{Enc}$. Formally, this is captured by the relation

$$\mathcal{R}_{\text{NY}} = \{(\mathbf{c}, \text{rnd}) : \exists m \in \mathcal{M}, \mathbf{c} = \text{tPKE}_0.\text{Enc}(\text{pk}_0, m; \text{rnd})\},$$

where the statement $\mathbf{c}$ is a ciphertext and the witness $\text{rnd} \in \mathcal{RD}$ is the randomness that produced it. Given a simulation-extractable NIZK $\text{NIZK} = (\text{Setup}, \text{Prove}, \text{Verify})$ for $\mathcal{R}_{\text{NY}}$, the transform yields the SimCCA -secure tPKE $\text{tPKE} = (\text{KeyGen}, \text{Enc}, \text{PartDec}, \text{FinDec})$ described in Fig. 5, where the CRS crs is treated as part of the public key and is available to all algorithms. We refer to [BKW26] for formal proofs of the security of this transform.

$\text{KeyGen}(1^\lambda, t, n)$	$\text{PartDec}(\text{sk}_i, \text{ct} = (\mathbf{c}, \pi))$
1 : $(\text{pk}_0, \text{sk}) \leftarrow \text{tPKE}_0.\text{KeyGen}(1^\lambda, t, n)$	1 : if $\text{NIZK}.\text{Verify}(\text{crs}, \mathbf{c}, \pi) = 0$ then
2 : $\text{crs} \leftarrow \text{NIZK}.\text{Setup}(1^\lambda)$	2 : return $\perp$
3 : return $(\text{pk} := (\text{pk}_0, \text{crs}), \text{sk})$	3 : $d_i \leftarrow \text{tPKE}_0.\text{PartDec}(\text{sk}_i, \mathbf{c})$
$\text{Enc}(\text{pk} = (\text{pk}_0, \text{crs}), m)$	4 : return d_i
1 : $\text{rnd} \xleftarrow{\$} \chi_{\text{rnd}}$	$\text{FinDec}(\text{ct} = (\mathbf{c}, \pi), \{d_i\}_{i \in \mathcal{I}})$
2 : $\mathbf{c} \leftarrow \text{tPKE}_0.\text{Enc}(\text{pk}_0, m; \text{rnd})$	$\text{// } \mathcal{I} \subseteq [n], \mathcal{I} = t$
3 : $\pi \leftarrow \text{NIZK}.\text{Prove}(\text{crs}, \mathbf{c}, \text{rnd})$	1 : return $\text{tPKE}_0.\text{FinDec}(\mathbf{c}, \{d_i\}_{i \in \mathcal{I}})$
4 : return $\text{ct} := (\mathbf{c}, \pi)$	

Fig. 5. A SimCCA -secure tPKE tPKE from a SemiMalSimCPA -secure tPKE tPKE_0 via the BKW transform [BKW26].

Remark 2. [BKW26] introduces a second transform for schemes that are only CPA-secure and do not support semi-malicious security (i.e., adversarially chosen randomness). In that case, their transform requires a non-interactive proof of randomness (NIPoR) to certify well-formedness and honest sampling of the encryption randomness. This, however, appears much more costly to instantiate.

5.3 CCA-secure threshold PKE

Finally, we discuss how to concretely instantiate a CCA-secure scheme from our tPKE scheme. We first outline our construction of a simulation extractable NIZK, which is the only missing ingredient in the BKW transform.

Simulation Extractable NIZK from Lattices. The Lantern proof system [LNP22] is a strong contender for instantiating our NIZK, as it excels at proving native lattice relationships and often provides the most compact sizes for small tasks.

Unfortunately, Lantern does not support witness extraction, and has furthermore only been proven to be weakly simulation-sound by Blömer et al. [BBP23] in the Random Oracle Model (ROM), meaning that while simulated proofs wouldn't help the adversary forge proofs for new statements, these simulated proofs may be malleable, which is unacceptable in our setting.

We propose two changes to upgrade Lantern to be simulation extractable, targeting the two above issues separately.

Witness Extractability. There exist several NIZK transformations to achieve straight-line extractability, including Fischlin's transform [Fis05], the (folklore) encryption-to-the-sky technique (see, e.g., [AKSY22] for a lattice-based instantiation), and Katsumata's transform [Kat21]. In this work, we use the encryption-to-the-sky paradigm since it is a black-box transform that does not require modifying the underlying NIZK system.

In this paradigm, the prover provides a public key encryption of the witness alongside the NIZK proof, while extending the proven relationship to ensure that the ciphertext correctly encrypts the witness. Assuming access to the PKE's secret key, one can easily extract the witness.

The simulation soundness of the NIZK is preserved by using a simulation-secure PKE (i.e. its ciphertexts can be simulated) whose public keys can be indistinguishably generated by a random oracle, ensuring that the adversary does not get access to the PKE's private key.

Strong Simulation Soundness. As explained above, Lantern only achieves *weak* simulation soundness, meaning simulated proofs may be malleable.

An early result by Lindell [Lin03] showed that one can generically upgrade an (adaptively) sound NIZK using a one-time signature and a specific commitment scheme. However, this transform requires to modify the statement proven, reducing practical efficiency.

Later on, several works [LPJY14, HKS24] leveraged a different approach to upgrade a weakly simulation sound NIZK relying on a (strongly unforgeable) one-time signature. They first sample and bind the signature public key to the proof when generating it, and then sign the proof to prevent malleability. The signature is then verified simultaneously with the proof, ensuring that anyone observing the proof cannot modify it since they would need to forge a new signature under the key chosen by the prover. These works differ in how the signature key is bound, [LPJY14] modifying their NIZK construction in depth, while [HKS24] focuses on NIZKs derived from a sigma protocol via the Fiat-Shamir transform, and simply adding the key to the hash input.

We adopt here the second, simpler and more efficient approach, though we revisit the formalism of [HKS24] by supporting any underlying *weak* simulation sound NIZK with labels, i.e. its proofs may be namespaced to specific labels, and

we further show that this technique can be combined with encryption-to-the-sky to achieve simulation extractability.

We formally define and prove these upgrades in Sec. B. We then instantiate it for Lantern, using a Regev public-key encryption for witness extraction, and ML-DSA as a strongly unforgeable signature scheme.

Our Scheme. By combining our tPKE scheme with the simulation-extractable NIZK above, we finally obtain a CCA-secure threshold PKE. Below, we briefly summarize the specifications and optimizations for our scheme.

In our CCA scheme, the encryptor needs to provide a NIZK proof that proves the ciphertext is generated from the encryption algorithm using a message from the plaintext space and encryption randomness drawn from the valid randomness space. As stated in the proof of Thm. 3, the randomness space for our semi-malicious CPA-secure tPKE is given by a closed two-norm ball with radius $e^{1/4}\sigma_r\sqrt{N(2k+1)}$. While this is a sufficient condition to prove the ciphertext well-formedness, we additionally assume that the randomness is confined within an infinity-norm ball with radius $B_r > 0$ to minimize the plaintext space for the witness encryption, hence a compact proof size. This bound B_r is chosen using Lem. 2, following from the fact that $\vec{r}$ is drawn from a Gaussian distribution.

For straight-line extractability, Regev-like public key encryptions of witness encryptions are also appended to the proof. Although there are many compact variants of Regev encryption, such as CRYSTALS-Kyber [BDK⁺18], the inconsistency in the ciphertext moduli of the tPKE and witness encryption may make language formation rather challenging and further increase the proof size. To avoid this issue, we use the same ciphertext modulus and, automatically, the same ciphertext dimension for both tPKE and witness encryptions. This not only helps with the proof size but also allows us to batch-encrypt several witnesses digit-by-digit into a single ciphertext rather than encrypting each witness separately. Observe that our tPKE should have a larger ciphertext modulus than conventional single-key Regev encryption, primarily to facilitate the flooding terms f_i , which asymptotically grow linearly to $\sqrt{Q}$. This ‘slack’ allows us to set the plaintext modulus large and encrypt several witnesses digit-by-digit. In the parameter selection section, we denote the number of witness encryptions as L . We also suggest encrypting them under different secret key/public key pairs, but with the same masking for all public keys. This is often referred to as the shared-A MLWE encryption.

Putting these discussions altogether, we can construct a concrete CCA-secure PKE scheme.

6 Evaluation

We now move on to the concrete instantiation of our construction.

6.1 B_λ and $B_{\lambda^{-1}}$

We first bound the bounds B_λ and the inverses $B_{\lambda^{-1}}$, as both the hint coefficients and the noise term contain these terms. While we have a mathematically proven bound for them in Lem. 5, there are some discrepancies compared to the real bounds, introducing extra overhead to the final ciphertext moduli.

Therefore, we instead use empirically computed bounds of $\gamma\lambda_i^T$ and $(\lambda_i^T)^{-1}$, following the analysis given in [KLL⁺26]. In [KLL⁺26], the authors reported that $\gamma\lambda_j^T$ is maximized when choosing the evaluation points as follows:

$$\mathcal{S} = \begin{cases} \{X^i\}_{-(t-1)/2 \leq i \leq (t-1)/2} & \text{if } t \text{ is odd} \\ \{X^i\}_{-t/2 < i \leq t/2} & \text{if } t \text{ is even} \end{cases}.$$

On the other hand, $(\lambda_j^T)^{-1}$ is maximized when the evaluation point set is given as follows:

$$\mathcal{S} = \begin{cases} \{0\} \cup \{X^i\}_{-(t-1)/2 \leq i < (t-1)/2} & \text{if } t \text{ is odd} \\ \{0\} \cup \{X^i\}_{-t/2 < i < t/2} & \text{if } t \text{ is even} \end{cases}.$$

We also use the well-known trick of using evaluation points over the subring $\mathbb{Z}[Y]/(Y^{N'} + 1)$ of $\mathcal{R} = \mathbb{Z}[X]/(X^N + 1)$, where $N' = 2^{\lceil \log n \rceil}/2$ and $Y = X^{N/N'}$. Then, the size of the Lagrange coefficients is only dependent on N' , rather than N . Based on this observation, we always choose the minimal subring for the Lagrange coefficients to minimize the parameters.

Recommended Parameters. Now, we choose and provide candidate parameters for our tPKE scheme and corresponding ciphertext sizes. We elaborate how to choose the Gaussian parameters in Sec. C. We fix the ring degree $N = 256$ and use the lattice estimator [APS15] to measure the security of the underlying hardness problems. In Table 3, we describe the recommended parameter sets based on the analysis above. In particular, we consider the following parameter regimes: maximum number of users $n \in \{16, 32\}$, threshold $t \in \{2, n/2, n-1\}$. Our proposed parameters achieve a 128-bit security level.

Ciphertext Sizes. Based on the recommended parameter sets, we compute the expected ciphertext sizes accordingly and summarize them in Table 4. For ML-DSA, we use Level-3 parameters. We also use the **LaZer** library to compute the expected size.

Currently, the size of CCA-secure PKE ciphertexts is dominated by the NIZK proof. We emphasize that the ciphertext size can be further reduced with optimizations in NIZK construction, as our security proof only requires the underlying NIZK scheme to be simulation extractable. We expect that a tailor-made construction of an simulation extractable NIZK system for our PKE scheme, rather than the current instantiation that relies on naïve transformations such as encryption-to-the-sky, could possibly reduce the NIZK size.

t	Q	q	k	σ_r	σ_s	σ_f	L
2	1	$2^{28.7}$	4	$2^{2.3}$	$2^{2.2}$	$2^{18.9}$	9
8	1	$2^{41.9}$	6	$2^{2.2}$	$2^{2.2}$	$2^{27.1}$	5
15	1	$2^{30.9}$	5	$2^{3.5}$	$2^{2.4}$	$2^{20.2}$	11
2	2^{32}	$2^{45.0}$	7	$2^{2.2}$	$2^{2.2}$	$2^{35.2}$	4
8	2^{32}	$2^{58.1}$	9	$2^{2.2}$	$2^{2.2}$	$2^{43.3}$	4
15	2^{32}	$2^{45.9}$	7	$2^{2.2}$	$2^{2.4}$	$2^{35.2}$	4
2	2^{60}	$2^{59.1}$	9	$2^{2.2}$	$2^{2.2}$	$2^{49.3}$	4
8	2^{60}	$2^{72.3}$	11	$2^{2.2}$	$2^{2.2}$	$2^{57.5}$	4
15	2^{60}	$2^{60.1}$	9	$2^{2.2}$	$2^{2.4}$	$2^{49.3}$	4

t	Q	q	k	σ_r	σ_s	σ_f	L
2	1	$2^{30.8}$	5	$2^{2.3}$	$2^{2.2}$	$2^{20.0}$	6
16	1	$2^{63.5}$	9	$2^{2.2}$	$2^{2.2}$	$2^{40.9}$	4
31	1	$2^{33.4}$	5	$2^{3.9}$	$2^{2.4}$	$2^{21.7}$	11
2	2^{32}	$2^{47.0}$	7	$2^{2.2}$	$2^{2.2}$	$2^{36.2}$	4
16	2^{32}	$2^{79.7}$	12	$2^{2.2}$	$2^{2.2}$	$2^{57.1}$	4
31	2^{32}	$2^{47.9}$	7	$2^{2.2}$	$2^{2.4}$	$2^{36.2}$	4
2	2^{60}	$2^{61.1}$	9	$2^{2.2}$	$2^{2.2}$	$2^{50.3}$	4
16	2^{60}	$2^{93.8}$	14	$2^{2.2}$	$2^{2.2}$	$2^{71.2}$	3
31	2^{60}	$2^{62.1}$	9	$2^{2.2}$	$2^{2.4}$	$2^{50.3}$	4

Table 3. Recommended parameter sets for (a) $n = 16$, (b) $n = 32$ parties. t , Q , q , k , σ_r , σ_s , and σ_f denote the threshold, the number of queries, ciphertext modulus, MLWE dimension, discrete Gaussian parameters for the encryption, key generation and partial decryption, respectively. L denotes the number of witness encryptions.

t	Q	CPA	CCA			
		ctxt	Sig	NIZK	WENC	ctxt
2	1	4.49	5.14	39.77	11.68	61.08
8	1	9.16	5.14	44.05	14.39	72.74
15	1	5.80	5.14	40.86	15.47	67.27
2	2^{32}	11.24	5.14	46.94	15.46	78.78
8	2^{32}	18.17	5.14	55.15	23.62	102.08
15	2^{32}	11.47	5.14	46.77	15.77	79.15
2	2^{60}	18.48	5.14	55.37	24.02	103.01
8	2^{60}	27.10	5.14	64.23	33.88	130.35
15	2^{60}	18.77	5.14	55.67	24.40	103.98

t	Q	CPA	CCA			
		ctxt	Sig	NIZK	WENC	ctxt
2	1	5.78	5.14	40.86	10.59	62.37
16	1	19.84	5.14	55.92	25.79	106.69
31	1	6.27	5.14	45.57	16.71	73.69
2	2^{32}	11.74	5.14	47.08	16.14	80.10
16	2^{32}	32.37	5.14	68.33	39.84	145.68
31	2^{32}	11.98	5.14	47.29	16.48	80.89
2	2^{60}	19.10	5.14	55.84	24.83	104.91
16	2^{60}	43.97	5.14	78.54	49.83	177.48
31	2^{60}	19.41	5.14	55.71	25.23	105.49

Table 4. The ciphertext sizes in KB when (a) $n = 16$, (b) $n = 32$ parties, based on the recommended parameter sets. For the CCA-secure variant, **Sig** denotes the size of the one-time signature for the strong simulation-soundness, and **NIZK** denotes the size of the zero-knowledge proof, and **WENC** denotes the size of the witness encryption for the straight-line extractability, respectively.

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A Deferred Proofs

A.1 Proof of Lem. 6

Proof. For any $\mathbf{x} \in \mathbb{Z}^m$, let $\mathcal{D}(\mathbf{x})$ be the probability mass function of the distribution $\{\mathbf{M}\mathbf{e} + \mathbf{f} \mid \mathbf{e} \leftarrow D_{\mathbb{Z}^n, \sigma}, \mathbf{f} \leftarrow D_{\mathbb{Z}^m, \sqrt{\Sigma}}\}$. We have:

$$\begin{aligned} \mathcal{D}(\mathbf{x}) &= \Pr[\mathbf{M}\mathbf{e} + \mathbf{f} = \mathbf{x} \mid \mathbf{e} \leftarrow D_{\mathbb{Z}^n, \sigma}, \mathbf{f} \leftarrow D_{\mathbb{Z}^m, \sqrt{\Sigma}}] \\ &= \sum_{\mathbf{y} \in \mathbb{Z}^n} D_{\mathbb{Z}^n, \sigma}(\mathbf{y}) \cdot D_{\mathbb{Z}^m, \sqrt{\Sigma}}(\mathbf{x} - \mathbf{M}\mathbf{y}) \\ &\propto \sum_{\mathbf{y} \in \mathbb{Z}^n} \exp \left[-\pi \left(\frac{1}{\sigma^2} \|\mathbf{y}\|^2 + (\mathbf{x} - \mathbf{M}\mathbf{y})^\top \Sigma^{-1} (\mathbf{x} - \mathbf{M}\mathbf{y}) \right) \right]. \end{aligned}$$

Then, we have

$$\begin{aligned} &\frac{1}{\sigma^2} \|\mathbf{y}\|^2 + (\mathbf{x} - \mathbf{M}\mathbf{y})^\top \Sigma^{-1} (\mathbf{x} - \mathbf{M}\mathbf{y}) \\ &= \mathbf{y}^\top \left(\frac{1}{\sigma^2} \mathbf{I}_n + \mathbf{M}^\top \Sigma^{-1} \mathbf{M} \right) \mathbf{y} - 2\mathbf{y}^\top (\mathbf{M}^\top \Sigma^{-1} \mathbf{x}) + \mathbf{x}^\top \Sigma^{-1} \mathbf{x} \\ &= (\mathbf{y} - \mathbf{c})^\top \Sigma_1^{-1} (\mathbf{y} - \mathbf{c}) + \mathbf{x}^\top (\Sigma^{-1} - \Sigma^{-1} \mathbf{M} \Sigma \mathbf{M}^\top \Sigma^{-1}) \mathbf{x} \\ &= (\mathbf{y} - \mathbf{c})^\top \Sigma_1^{-1} (\mathbf{y} - \mathbf{c}) + \mathbf{x}^\top (\sigma^2 \mathbf{M} \mathbf{M}^\top + \Sigma)^{-1} \mathbf{x}, \end{aligned}$$

where $\Sigma_1^{-1} := \frac{1}{\sigma^2} \mathbf{I}_n + \mathbf{M}^\top \Sigma^{-1} \mathbf{M}$, $\mathbf{c} := \Sigma_1 \mathbf{M}^\top \Sigma^{-1} \mathbf{x}$ and the last simplification is derived from the Woodbury Matrix Identity [Dun44]. This implies that

$$\mathcal{D}(\mathbf{x}) \propto \rho_{\sqrt{\sigma^2 \mathbf{M} \mathbf{M}^\top + \Sigma}}(\mathbf{x}) \cdot \sum_{\mathbf{y} \in \mathbb{Z}^n} \rho_{\sqrt{\Sigma_1}}(\mathbf{y} - \mathbf{c}).$$

From the condition in the lemma, we check the smoothing parameter requirement. The smallest singular value of $\sqrt{\Sigma_1^{-1}}$ satisfies:

$$s_{\min}(\sqrt{\Sigma_1^{-1}}) = \sqrt{\lambda_{\min}(\Sigma_1^{-1})} = \sqrt{\lambda_{\min} \left(\frac{1}{\sigma^2} \mathbf{I}_n + \mathbf{M}^\top \Sigma^{-1} \mathbf{M} \right)},$$

where $s_{\min}(\cdot)$ denotes the smallest singular value and $\lambda_{\min}(\cdot)$ denotes the smallest eigenvalue. Given $\frac{1}{\sigma^2} + \|\mathbf{M}\sqrt{\Sigma^{-1}}\|_2^2 \leq \frac{1}{\eta_\varepsilon(\mathbb{Z}^n)^2}$ and noting that $\|\mathbf{M}\sqrt{\Sigma^{-1}}\|_2^2$ is the largest eigenvalue of $\mathbf{M}^\top \Sigma^{-1} \mathbf{M}$, this ensures the covariance Σ_1 is "wide" enough such that for any offset $\mathbf{c}$, the sum over the lattice $\mathbb{Z}^n$ is nearly constant:

$$\sum_{\mathbf{y} \in \mathbb{Z}^n} \rho_{\sqrt{\Sigma_1}}(\mathbf{y} - \mathbf{c}) \in \left[\frac{1 - \varepsilon}{1 + \varepsilon}, 1 \right] \rho_{\sqrt{\Sigma_1}}(\mathbb{Z}^n).$$

due to [AGHS13, Lemma 4]. This leads to the bound:

$$\mathcal{D}(\mathbf{x}) \in [1 - 4\varepsilon, 1 + 4\varepsilon] \cdot D_{\mathbb{Z}^m, \sqrt{\sigma^2 \mathbf{M} \mathbf{M}^\top + \Sigma}}(\mathbf{x}),$$

which concludes that the statistical distance is bounded by 4ε . $\square$

A.2 Proof of Lem. 7

Proof. Consider an input $\mathbf{x} + \mathbf{y}$. Then the algorithm $\text{Rerand}(\mathbf{M}, \mathbf{x} + \mathbf{y}, \sigma, s)$ computes the value $\mathbf{M}(\mathbf{x} + \mathbf{y}) + \mathbf{f}$ where $\mathbf{f} \leftarrow D_{\mathbb{Z}^n, \sqrt{\Sigma}}$ for $\Sigma := \sigma^2 s^2 \mathbf{I}_m - \sigma^2 \mathbf{M} \mathbf{M}^\top$. Note that Σ is a positive semidefinite matrix since

$$\|\mathbf{M}\|_2 \leq \sqrt{1 - \eta_\epsilon^2(\mathbb{Z}^m)/\sigma^2} \cdot s \leq \sigma s.$$

Then, Lem. 6 states that the output of $\text{Rerand}(\mathbf{M}, \mathbf{x} + \mathbf{y}, \sigma, s)$ is statistically close to $D_{\mathbb{Z}^m, \sigma s}$ as long as $\frac{1}{\sigma^2} + \|\mathbf{M}\sqrt{\Sigma^{-1}}\|_2^2 \leq \frac{1}{\eta_\epsilon^2(\mathbb{Z}^n)}$. We claim that this inequality holds given the conditions. Note that the smallest eigenvalue of Σ is at least

$$s^2 \sigma^2 - \sigma^2 \|\mathbf{M}\|_2^2 \geq \frac{\eta_\epsilon^2(\mathbb{Z}^n) \sigma^2}{\sigma^2 - \eta_\epsilon^2(\mathbb{Z}^n)} \|\mathbf{M}\|_2^2.$$

This directly follows from the condition $\|\mathbf{M}\|_2 \leq \sqrt{\frac{\sigma^2 - \eta_\epsilon^2(\mathbb{Z}^n)}{\sigma^2}} \cdot s$. According to [KLSS23, Lemma 5], this implies $\|\Sigma^{-1}\|_2^2 \leq \frac{\sigma^2 - \eta_\epsilon^2(\mathbb{Z}^n)}{\eta_\epsilon^2(\mathbb{Z}^n) \sigma^2} \|\mathbf{M}\|_2^{-2}$. Then, from the property of the matrix two-norm,

$$\|\mathbf{M}\sqrt{\Sigma^{-1}}\|_2^2 \leq \|\mathbf{M}\|_2^2 \cdot \|\sqrt{\Sigma^{-1}}\|_2 \leq \frac{\sigma^2 - \eta_\epsilon^2(\mathbb{Z}^n)}{\eta_\epsilon^2(\mathbb{Z}^n) \sigma^2} = \frac{1}{\eta_\epsilon^2(\mathbb{Z}^n)} - \frac{1}{\sigma^2},$$

thereby showing that the claim is true and hence the algorithm outputs the correct value. $\square$

B Constructing a Simulation Extractable NIZK from a Weak Simulation Secure NIZK

This section formalizes the transform from a weak simulation sound labelled NIZK to a simulation extractable NIZK, using the encryption-to-the-sky technique for extractability, and a strongly unforgeable one-time signature for strong simulation soundness.

B.1 Building Blocks

The transform combines three components, which we define below: a strongly unforgeable one-time signature (Def. 13), a labelled weakly simulation sound NIZK (Defs. 14 and 15), and a simulation-secure PKE whose public keys can themselves be simulated (Def. 17). Throughout, labels are arbitrary bit strings $\ell \in \{0, 1\}^*$, and we write $x \in \mathcal{L}$ (resp. $x \notin \mathcal{L}$) to mean that a witness w with $(x, w) \in \mathcal{R}$ exists (resp. does not).

Strongly Unforgeable One-Time Signature.

Definition 12 (One-Time Signature). A one-time signature scheme is a tuple $\text{OTS} = (\text{KeyGen}, \text{Sign}, \text{Verify})$ with the following syntax:

- $\text{KeyGen}(1^\lambda) \rightarrow (\text{vk}, \text{sk})$ Generates a verification key and a signing key.
- $\text{Sign}(\text{sk}, m) \rightarrow \sigma$ Given a signing key sk and a message $m \in \{0, 1\}^*$, outputs a signature σ .
- $\text{Verify}(\text{vk}, m, \sigma) \rightarrow b$ Given a verification key vk , a message m , and a signature σ , outputs a verdict $b \in \{0, 1\}$.

It is correct if for all $(\text{vk}, \text{sk}) \leftarrow \text{KeyGen}(1^\lambda)$ and all $m \in \{0, 1\}^*$, $\text{Verify}(\text{vk}, m, \text{Sign}(\text{sk}, m)) = 1$.

Definition 13 (Strong One-Time Unforgeability). A one-time signature $\text{OTS} = (\text{KeyGen}, \text{Sign}, \text{Verify})$ is strongly unforgeable under a one-time chosen-message attack if

$$\Pr[\text{sOTS}_{\text{OTS}, \mathcal{A}}(1^\lambda) = 1] \leq \text{negl}(\lambda)$$

for every PPT adversary $\mathcal{A}$, where the experiment $\text{sOTS}_{\text{OTS}, \mathcal{A}}$ is defined in Fig. 6. The adversary may query the signing oracle at most once.

$\text{sOTS}_{\text{OTS}, \mathcal{A}}(1^\lambda)$	$\text{OSign}(m)$
1 : $(\text{vk}, \text{sk}) \leftarrow \text{KeyGen}(1^\lambda)$	1 : assert $\mathcal{Q} = \emptyset$ // one-time
2 : $\mathcal{Q} \leftarrow \emptyset$	2 : $\sigma \leftarrow \text{Sign}(\text{sk}, m)$
3 : $(m^*, \sigma^*) \leftarrow \mathcal{A}^{\text{OSign}}(\text{vk})$	3 : $\mathcal{Q} \leftarrow \mathcal{Q} \cup \{(m, \sigma)\}$
4 : return $\text{Verify}(\text{vk}, m^*, \sigma^*) = 1$	4 : return σ
5 : $\wedge (m^*, \sigma^*) \notin \mathcal{Q}$	

Fig. 6. Experiment for the strong one-time unforgeability of a signature scheme.

Labelled Weakly Simulation-Sound NIZK.

Definition 14 (Labelled NIZK). A labelled non-interactive zero-knowledge proof system for a language $\mathcal{L}$ with associated relation $\mathcal{R}$ is a tuple $\text{NIZK} = (\text{Setup}, \text{Prove}, \text{Verify})$ with the following syntax:

- $\text{Setup}(1^\lambda) \rightarrow \text{crs}$ Generates a common reference string.
- $\text{Prove}(\text{crs}, x, \ell, w) \rightarrow \pi$ Given a CRS crs , a label $\ell \in \{0, 1\}^*$, and $(x, w) \in \mathcal{R}$, outputs a proof π .
- $\text{Verify}(\text{crs}, x, \ell, \pi) \rightarrow b$ Given a CRS crs , a statement x , a label ℓ , and a proof π , outputs a verdict $b \in \{0, 1\}$.

It is perfectly correct if for all $(x, w) \in \mathcal{R}$, all labels ℓ , and all $\text{crs} \leftarrow \text{Setup}(1^\lambda)$, $\text{Verify}(\text{crs}, x, \ell, \pi) = 1$ where $\pi \leftarrow \text{Prove}(\text{crs}, x, \ell, w)$.

Definition 15 (Weak Simulation Soundness). A labelled NIZK $\text{NIZK} = (\text{Setup}, \text{Prove}, \text{Verify})$ for a relation $\mathcal{R}$ is weakly simulation-sound if there exists a PPT simulator $\text{Sim} = (\text{Sim.Setup}, \text{Sim.Sim})$ such that the following two properties hold, for the experiments defined in Fig. 7.

Zero-knowledge: For every PPT adversary $\mathcal{A}$,

$$|\Pr[\text{ZK}_{\text{NIZK}, \text{Sim}, \mathcal{A}}^0(1^\lambda) = 1] - \Pr[\text{ZK}_{\text{NIZK}, \text{Sim}, \mathcal{A}}^1(1^\lambda) = 1]| \leq \text{negl}(\lambda),$$

i.e., a simulated CRS together with simulated proofs (which are produced without a witness) is indistinguishable from an honest CRS with honest proofs.

Weak simulation-soundness: For every PPT adversary $\mathcal{A}$,

$$\Pr[\text{WSS}_{\text{NIZK}, \text{Sim}, \mathcal{A}}(1^\lambda) = 1] \leq \text{negl}(\lambda),$$

i.e., even after observing simulated proofs, $\mathcal{A}$ cannot produce a valid proof for a false statement under a fresh statement-label pair.

$\text{ZK}_{\text{NIZK}, \text{Sim}, \mathcal{A}}^b(1^\lambda)$	$\text{WSS}_{\text{NIZK}, \text{Sim}, \mathcal{A}}(1^\lambda)$
1 : if $b = 0$ then $\text{crs} \xleftarrow{\$} \text{Setup}(1^\lambda)$	1 : $(\text{crs}, \text{st}) \xleftarrow{\$} \text{Sim.Setup}(1^\lambda)$
2 : if $b = 1$ then $(\text{crs}, \text{st}) \xleftarrow{\$} \text{Sim.Setup}(1^\lambda)$	2 : $\mathcal{Q} \leftarrow \emptyset$
3 : return $\mathcal{A}^{\text{ProveO}}(\text{crs})$	3 : $(x^*, \ell^*, \pi^*) \leftarrow \mathcal{A}^{\text{SimO}}(\text{crs})$
$\text{ProveO}(x, \ell, w)$	4 : return $\text{Verify}(\text{crs}, x^*, \ell^*, \pi^*) = 1$
1 : assert $(x, w) \in \mathcal{R}$	5 : $\wedge (x^*, \ell^*) \notin \mathcal{Q} \wedge x^* \notin \mathcal{L}$
2 : if $b = 0$ then $\pi \xleftarrow{\$} \text{Prove}(\text{crs}, x, \ell, w)$	$\text{SimO}(x, \ell)$
3 : if $b = 1$ then $(\pi, \text{st}) \xleftarrow{\$} \text{Sim.Sim}(x, \ell, \text{st})$	1 : $(\pi, \text{st}) \xleftarrow{\$} \text{Sim.Sim}(x, \ell, \text{st})$
4 : return π	2 : $\mathcal{Q} \leftarrow \mathcal{Q} \cup \{(x, \ell)\}$
	3 : return π

Fig. 7. Experiments for the weak simulation security of a labelled NIZK: zero-knowledge (ZK^b) and weak simulation-soundness (WSS).

Simulation-Secure PKE with Uniform Keys.

Definition 16 (PKE). A public-key encryption scheme is a tuple $\text{PKE} = (\text{KeyGen}, \text{Enc}, \text{Dec})$ with public-key space $\mathcal{PK}$ and encryption randomness space $\mathcal{RD}$, with the following syntax:

- $\text{KeyGen}(1^\lambda) \rightarrow (\text{pk}, \text{sk})$ Generates a public key $\text{pk} \in \mathcal{PK}$ and a secret key.
- $\text{Enc}(\text{pk}, m) \rightarrow \mathbf{c}$ Encrypts a message m under pk , producing a ciphertext $\mathbf{c}$. We write $\text{Enc}(\text{pk}, m; \text{rnd})$ for the deterministic encryption using explicit randomness $\text{rnd} \in \mathcal{RD}$.

- $\text{Dec}(\text{sk}, \mathbf{c}) \rightarrow m$ Decrypts $\mathbf{c}$ using sk .

It is correct over $\mathcal{RD}$ if, with overwhelming probability over $(\text{pk}, \text{sk}) \leftarrow \text{KeyGen}(1^\lambda)$, decryption succeeds for every accepted randomness simultaneously:

$$\Pr_{(\text{pk}, \text{sk}) \leftarrow \text{KeyGen}(1^\lambda)} [\exists m, \text{rnd} \in \mathcal{RD} : \text{Dec}(\text{sk}, \text{Enc}(\text{pk}, m; \text{rnd})) \neq m] \leq \text{negl}(\lambda).$$

In particular, decryption recovers the message even for an adversarially chosen $\text{rnd} \in \mathcal{RD}$, not only for honestly sampled randomness.

Definition 17 (Simulation-Secure PKE with Uniform Keys). A PKE scheme $\text{PKE} = (\text{KeyGen}, \text{Enc}, \text{Dec})$ with public-key space $\mathcal{PK}$ is simulation-secure with uniform keys if there exists a PPT ciphertext simulator Sim.Enc such that the following two properties hold. We call a key sampled uniformly from $\mathcal{PK}$ an oblivious key, i.e. it carries no secret.

Key simulation: A key produced by the honest key generation is indistinguishable from a uniformly sampled one:

$$\{\text{pk} : (\text{pk}, \text{sk}) \leftarrow \text{KeyGen}(1^\lambda)\} \approx_c \{\text{pk} : \text{pk} \xleftarrow{\$} \mathcal{PK}\}.$$

Ciphertext simulation: Under an oblivious key, ciphertexts can be produced without the message: for every PPT adversary $\mathcal{A}$,

$$|\Pr[\text{CS}_{\text{PKE}, \mathcal{A}}^0(1^\lambda) = 1] - \Pr[\text{CS}_{\text{PKE}, \mathcal{A}}^1(1^\lambda) = 1]| \leq \text{negl}(\lambda),$$

where the experiment $\text{CS}_{\text{PKE}, \mathcal{A}}^b$ is defined in Fig. 8.

$\text{CS}_{\text{PKE}, \mathcal{A}}^b(1^\lambda)$	$\text{EncO}(m)$
1 : $\text{pk} \xleftarrow{\$} \mathcal{PK}$	1 : if $b = 0$ then $\mathbf{c} \xleftarrow{\$} \text{Enc}(\text{pk}, m)$
2 : return $\mathcal{A}^{\text{EncO}}(\text{pk})$	2 : if $b = 1$ then $\mathbf{c} \xleftarrow{\$} \text{Sim.Enc}(\text{pk})$
	3 : return $\mathbf{c}$

Fig. 8. Experiment for the ciphertext simulation of a PKE scheme under an oblivious (uniformly sampled) public key.

B.2 The Transform

Fix a relation $\mathcal{R}$ with language $\mathcal{L}$. The transform combines the three building blocks:

- a strongly unforgeable one-time signature $\text{OTS} = (\text{KeyGen}, \text{Sign}, \text{Verify})$ (Def. 13);

- a simulation-secure PKE $\text{PKE} = (\text{KeyGen}, \text{Enc}, \text{Dec})$ with uniform keys (Def. 17), public-key space $\mathcal{PK}$, ciphertext simulator PKE.Sim.Enc , and encryption randomness space $\mathcal{RD}$ over which it is correct (Def. 16);
- a labelled weakly simulation-sound NIZK $\text{NIZK}_0 = (\text{Setup}, \text{Prove}, \text{Verify})$ (Def. 15), with simulator $\text{Sim}_0 = (\text{Sim}_0.\text{Setup}, \text{Sim}_0.\text{Sim})$, for the augmented relation

$$\mathcal{R}_0 = \left\{ ((x, \text{pk}, \text{c}), (w, \text{rnd})) : \begin{array}{l} (x, w) \in \mathcal{R} \wedge \text{rnd} \in \mathcal{RD} \\ \wedge \text{c} = \text{PKE.Enc}(\text{pk}, w; \text{rnd}) \end{array} \right\},$$

with language $\mathcal{L}_0$.

The resulting NIZK $\text{NIZK} = (\text{Setup}, \text{Prove}, \text{Verify})$ for $\mathcal{R}$ is described in Fig. 9. It is defined in the random oracle model, using a hash function $H : \{0, 1\}^* \rightarrow \mathcal{PK}$ modeled as a random oracle to derive the PKE public key $\text{pk} = H(\text{seed})$ from a public seed placed in the CRS; as H is a random oracle, this key is an oblivious (uniform) key that no party can invert to a trapdoor. Intuitively, each proof encrypts the witness under pk , uses NIZK_0 to prove that this ciphertext encrypts a valid witness, and binds the whole proof to a freshly generated one-time key vk , which serves both as the NIZK_0 label and as the key under which the proof is signed.

Setup(1^λ)	Prove($\text{crs} = (\text{crs}_0, \text{seed}), x, w$)
1 : $\text{crs}_0 \leftarrow \text{NIZK}_0.\text{Setup}(1^\lambda)$	1 : $\text{pk} := H(\text{seed})$
2 : $\text{seed} \xleftarrow{\$} \{0, 1\}^\lambda$	2 : $(\text{vk}, \text{sk}) \leftarrow \text{OTS.KeyGen}(1^\lambda)$
3 : return $\text{crs} := (\text{crs}_0, \text{seed})$	3 : $(\text{c}, \text{rnd}) \leftarrow \text{PKE.Enc}(\text{pk}, w)$
Verify($\text{crs} = (\text{crs}_0, \text{seed}), x, \pi = (\text{vk}, \text{c}, \pi_0, \sigma)$)	4 : $\pi_0 \leftarrow \text{NIZK}_0.\text{Prove}(\text{crs}_0, (x, \text{pk}, \text{c}), \text{vk}, (w, \text{rnd}))$
1 : $\text{pk} := H(\text{seed})$	5 : $\sigma \leftarrow \text{OTS.Sign}(\text{sk}, (x, \text{c}, \pi_0))$
2 : $b_1 \leftarrow \text{OTS.Verify}(\text{vk}, (x, \text{c}, \pi_0), \sigma)$	6 : return $\pi := (\text{vk}, \text{c}, \pi_0, \sigma)$
3 : $b_2 \leftarrow \text{NIZK}_0.\text{Verify}(\text{crs}_0, (x, \text{pk}, \text{c}), \text{vk}, \pi_0)$	
4 : return $b_1 \wedge b_2$	

Fig. 9. The simulation-extractable NIZK NIZK from a labelled weakly simulation-sound NIZK NIZK_0 , a strongly unforgeable one-time signature OTS , and a simulation-secure PKE PKE .

Theorem 4. *In the random-oracle model, if NIZK_0 is a labelled weakly simulation-sound NIZK for $\mathcal{R}_0$, OTS is strongly one-time unforgeable, and PKE is simulation-secure with uniform keys, then the NIZK NIZK of Fig. 9 is simulation-extractable (Def. 11) for $\mathcal{R}$, with the simulator Sim of Fig. 10.*

Proof. We show that SIMEXT^0 and SIMEXT^1 (Fig. 4) are indistinguishable through a sequence of hybrids. Fix a PPT adversary $\mathcal{A}$ making at most q queries to ProveO .

Sim.Setup(1^λ)	Sim.Sim($x, st = (st_0, pk, sk_{PKE})$)
1 : $(crs_0, st_0) \leftarrow \text{NIZK}_0.\text{Sim.Setup}(1^\lambda)$	1 : $(vk, sk) \leftarrow \text{OTS.KeyGen}(1^\lambda)$
2 : $(pk, sk_{PKE}) \leftarrow \text{PKE.KeyGen}(1^\lambda)$	2 : $c \leftarrow \text{PKE.Sim.Enc}(pk)$
3 : $seed \xleftarrow{\$} \{0, 1\}^\lambda$	3 : $(\pi_0, st_0) \leftarrow \text{NIZK}_0.\text{Sim.Sim}((x, pk, c), vk, st_0)$
4 : $H[seed] := pk$ // program the oracle	4 : $\sigma \leftarrow \text{OTS.Sign}(sk, (x, c, \pi_0))$
5 : return $(crs := (crs_0, seed), st := (st_0, pk, sk_{PKE}))$	5 : return $(\pi := (vk, c, \pi_0, \sigma), st)$
	Sim.Ext($x, \pi = (vk, c, \pi_0, \sigma), st$)
	1 : return $w := \text{PKE.Dec}(sk_{PKE}, c)$

Fig. 10. The simulator $\text{Sim} = (\text{Sim.Setup}, \text{Sim.Sim}, \text{Sim.Ext})$ for the NIZK of Fig. 9.

H_0 : the real experiment SIMEXT^0 . Here $crs = (crs_0, seed)$ with $crs_0 \leftarrow \text{NIZK}_0.\text{Setup}(1^\lambda)$ and $seed \xleftarrow{\$} \{0, 1\}^\lambda$; the PKE key $pk = H(seed)$ is uniform over $\mathcal{PK}$, since H is a random oracle and $seed$ is fresh. The oracle ProveO answers with honest proofs of Fig. 9, and VerifyO returns $\text{Verify}(crs, x, \pi)$.

H_1 : as H_0 , but we simulate the base CRS and every base proof: $(crs_0, st_0) \leftarrow \text{NIZK}_0.\text{Sim.Setup}(1^\lambda)$, and each ProveO call produces $\pi_0 \leftarrow \text{NIZK}_0.\text{Sim.Sim}((x, pk, c), vk, st_0)$ rather than an honest proof (note that $c \leftarrow \text{PKE.Enc}(pk, w)$ still encrypts the real witness). Since $(x, w) \in \mathcal{R}$ and $\text{rnd} \in \mathcal{RD}$ imply $((x, pk, c), (w, \text{rnd})) \in \mathcal{R}_0$, this is a legitimate use of the zero-knowledge oracle.

We can define an adversary $\mathcal{B}_1$ against the zero-knowledge of NIZK_0 that simulates H_0 and H_1 for $\mathcal{A}$. It uses the crs and ProveO oracle of its zero-knowledge challenger to answer $\mathcal{A}$'s queries, and outputs whatever $\mathcal{A}$ outputs. Then,

$$|\Pr[H_1 = 1] - \Pr[H_0 = 1]| = \text{Adv}_{\text{NIZK}_0}^{\text{zk}}(\mathcal{B}_1),$$

H_2 : as H_1 , but each ProveO ciphertext is simulated, $c \leftarrow \text{PKE.Sim.Enc}(pk)$ instead of $\text{PKE.Enc}(pk, w)$. The oracle no longer uses w and now coincides with Sim.Sim from Fig. 10.

By the simulation security of PKE with uniform keys, we can define an adversary $\mathcal{B}_2$ against the ciphertext simulation of PKE such that $|\Pr[H_2 = 1] - \Pr[H_1 = 1]| = \text{Adv}_{\text{PKE}}^{\text{cs}}(\mathcal{B}_2)$.

H_3 : as H_2 , but $pk = H(seed)$ is now *trapdoored*: we sample $(pk, sk_{PKE}) \leftarrow \text{PKE.KeyGen}(1^\lambda)$ and program $H[seed] := pk$, exactly as Sim.Setup does.

Since $\mathcal{A}$ makes no random-oracle queries before receiving crs , distinguishing this programming from a uniform pk directly breaks the key simulation of PKE. Formally, we can define an adversary $\mathcal{B}_3$ against the key simulation of PKE such that $|\Pr[H_3 = 1] - \Pr[H_2 = 1]| = \text{Adv}_{\text{PKE}}^{\text{ks}}(\mathcal{B}_3)$.

At this point, crs and ProveO are exactly those of Sim.Setup and Sim.Sim , while VerifyO still returns $\text{Verify}(crs, x, \pi)$.

H_4 : the ideal experiment SIMEXT^1 , differing from H_3 only in VerifyO : on a query (x, π) with $\text{Verify}(crs, x, \pi) = 1$ and $(x, \pi) \notin \mathcal{W}$, it extracts $w' \leftarrow \text{PKE.Dec}(sk_{PKE}, c)$ and reports an extraction failure when $(x, w') \notin \mathcal{R}$ instead of returning 1. The

two hybrids agree unless $\mathcal{A}$ makes a `VerifyO` query $(x, \pi = (\mathbf{vk}, \mathbf{c}, \pi_0, \sigma))$ triggering the following *bad event* `Bad`:

1. $\text{OTS.Verify}(\mathbf{vk}, (x, \mathbf{c}, \pi_0), \sigma) = 1$ and $\text{NIZK}_0.\text{Verify}(\text{crs}_0, (x, \mathbf{pk}, \mathbf{c}), \mathbf{vk}, \pi_0) = 1$ (the proof verifies);
2. $(x, \pi) \notin \mathcal{W}$; and
3. $(x, w') \notin \mathcal{R}$ for $w' \leftarrow \text{PKE.Dec}(\text{sk}_{\text{PKE}}, \mathbf{c})$.

It remains to bound $\Pr[\text{Bad}]$. Let $\mathbf{vk}_1, \dots, \mathbf{vk}_q$ be the one-time keys sampled across the simulated proofs, and for the i -th one let x_i be the corresponding statement and $(\mathbf{vk}_i, \mathbf{c}_i, \pi_{0,i}, \sigma_i) \in \mathcal{W}$ the corresponding proof. We distinguish two cases according to the label $\mathbf{vk}$ of the bad proof.

Case 1: $\mathbf{vk} \notin \{\mathbf{vk}_1, \dots, \mathbf{vk}_q\}$. Then $\mathbf{vk}$ was never used as a label by $\text{Sim}_0.\text{Sim}$, so $((x, \mathbf{pk}, \mathbf{c}), \mathbf{vk})$ is a fresh statement-label pair. Moreover $(x, \mathbf{pk}, \mathbf{c}) \notin \mathcal{L}_0$: otherwise $\mathbf{c} = \text{PKE.Enc}(\mathbf{pk}, w; \text{rnd})$ for some $(x, w) \in \mathcal{R}$ and $\text{rnd} \in \mathcal{RD}$, and correctness of $(\mathbf{pk}, \text{sk}_{\text{PKE}})$ over $\mathcal{RD}$ (Def. 16, failing with probability at most δ_{PKE}) would give $w' = w$, contradicting condition 3. Hence π_0 is a valid NIZK_0 proof for a false statement under a fresh pair, and the reduction $\mathcal{B}_4$ breaks weak simulation-soundness. This case contributes at most $\text{Adv}_{\text{NIZK}_0}^{\text{wss}}(\mathcal{B}_4) + \delta_{\text{PKE}}$.

Case 2: $\mathbf{vk} = \mathbf{vk}_i$ for some i . Since $(x, \pi) \notin \mathcal{W}$, we have $(x, \mathbf{c}, \pi_0, \sigma) \neq (x_i, \mathbf{c}_i, \pi_{0,i}, \sigma_i)$, so $((x, \mathbf{c}, \pi_0), \sigma)$ is either a signature on a message never signed under $\mathbf{vk}_i$ or a new signature on the already-signed $(x_i, \mathbf{c}_i, \pi_{0,i})$ —in both cases a fresh forgery under $\mathbf{vk}_i$. The reduction $\mathcal{B}_5$ sets up $\mathbf{pk}$ as in H_3 , guesses $i \in [q]$, embeds its challenge key as $\mathbf{vk}_i$, and uses its single signing query to produce σ_i ; this case contributes at most $q \cdot \text{Adv}_{\text{OTS}}^{\text{OTS}}(\mathcal{B}_5)$.

Therefore $\Pr[\text{Bad}] \leq q \cdot \text{Adv}_{\text{OTS}}^{\text{OTS}}(\mathcal{B}_5) + \text{Adv}_{\text{NIZK}_0}^{\text{wss}}(\mathcal{B}_4) + \delta_{\text{PKE}}$, and since H_3 and H_4 agree unless `Bad` occurs, $|\Pr[\text{H}_4 = 1] - \Pr[\text{H}_3 = 1]| \leq \Pr[\text{Bad}]$. Chaining the four transitions and using $\text{H}_0 = \text{SIMEXT}^0$ and $\text{H}_4 = \text{SIMEXT}^1$ yields

$$\begin{aligned} |\Pr[\text{SIMEXT}^0 = 1] - \Pr[\text{SIMEXT}^1 = 1]| &\leq \text{Adv}_{\text{NIZK}_0}^{\text{zk}}(\mathcal{B}_1) + \text{Adv}_{\text{PKE}}^{\text{cs}}(\mathcal{B}_2) + \text{Adv}_{\text{PKE}}^{\text{ks}}(\mathcal{B}_3) \\ &\quad + \text{Adv}_{\text{NIZK}_0}^{\text{wss}}(\mathcal{B}_4) + q \cdot \text{Adv}_{\text{OTS}}^{\text{OTS}}(\mathcal{B}_5) + \delta_{\text{PKE}}, \end{aligned}$$

which is negligible under the theorem's assumptions and concludes the proof. $\square$

C Parameter Selection

We elaborate on how to choose parameters $\mathbf{pp} = (p, q, N, k, \sigma_s, B_r, \sigma_f)$ for our tPKE scheme and then provide a recommended parameter set and corresponding ciphertext sizes. First, we begin by explaining how to choose the (adaptive) hint-MLWE parameters. Recall that Thm. 1 states that the problem $\text{AHMLWE}_{q,k,\ell,Q,\sigma_{\bar{x}},\sigma_y,\mathcal{V}_B}$ reduces to $\text{MLWE}_{q,k,\ell,\sigma}$ where σ is a positive real number defined as $\sigma^{-2} =$

$2(\sigma_{\bar{x}}^{-2} + \kappa^2 Q \sigma_y^{-2})$ for a constant $\kappa := \sqrt{\frac{1 - \sqrt{1 - 2B\eta_{\epsilon}^2(\mathbb{Z}^N)/(\pi\sigma_y^2)}}{\eta_{\epsilon}^2(\mathbb{Z}^N)/(\pi\sigma_y^2)}}$. Note that we simply chose the hint set to be $\mathcal{V}_B$, so we do not need to consider the first condition in the theorem. Next, following the convention for MLWE security, we fix

$\sigma = 3.2$. Note that this will naturally satisfy the third condition required for the reduction; $\sigma \geq \eta_\epsilon(\mathbb{Z}^N)/\sqrt{\pi}$ for practical parameters with $N \leq 2^{16}$. We also observe that κ is essentially a monotonically decreasing function with respect to σ_y , which outputs $\sqrt{2B}$ if σ_y has the smallest possible value, $\sqrt{2B/\pi} \cdot \eta_\epsilon(\mathbb{Z}^N)$, and outputs $\approx \sqrt{B}$ if σ_y is sufficiently large. Since having κ larger than the real value only makes the value of σ_y increase by less than half a bit, which has no negative effect on security at all, we can simply set $\kappa = \sqrt{2B}$. Then, the final condition reduces to $\sigma^{-2} = 2(\sigma_{\bar{x}}^{-2} + 2BQ\sigma_y^{-2})$. Since $\sigma_{\bar{x}}$ is real, we have $\sigma^{-2} > 4BQ\sigma_y^{-2}$. Note that this implies $\sigma_y > 2\sqrt{BQ/\pi} \cdot \eta_\epsilon(\mathbb{Z}^N)$ combined with the third condition, thereby satisfying the condition $\sigma_y \geq \sqrt{2B/\pi} \cdot \eta_\epsilon(\mathbb{Z}^N)$ naturally. Consequently, we only need to consider the final condition $\sigma^{-2} = 2(\sigma_{\bar{x}}^{-2} + 2BQ\sigma_y^{-2})$ when choosing the parameters for the adaptive Hint-MLWE problem.

Now, for our scheme to be correct and secure, it needs to satisfy Thm. 2 and Thm. 3. Given the threshold and the number of parties (t, n) , these theorems state that the following conditions need to be satisfied:

1. $\gamma(\rho_r\sigma_r + e^{1/2} \cdot 2kN\sigma_r\sigma_s) + tB_\lambda\rho_f\sigma_f < \frac{q}{2p} - \frac{p}{2}$,
2. $\text{AHMLWE}_{q,k,k,Q,\sigma_s,\sigma_f,\mathcal{V}_{B_1}}$ is hard for $B_1 = e^{1/2}N^2(2k+1)(B_{\lambda^{-1}}\sigma_r)^2$,
3. $\text{AHMLWE}_{q,k+1,k,n,\sigma_r,\sigma_f,\mathcal{V}_{B_2}}$ is hard for $B_2 = B_{\lambda^{-1}}^2(1 + 2e^{1/2}kN^2\sigma_s^2)$,
4. $\text{AHMLWE}_{q,k+Q_c,k,Q,\sigma_1,\sigma_f,\mathcal{V}_{B_3}}$ is hard for $\sigma_1^2 = \frac{\sigma_f^2}{2\gamma^2} \left(1 + \sqrt{1 - \frac{2\gamma^2\eta_\epsilon(\mathbb{Z}^N)^2}{\pi\sigma_f^2}}\right)$ and $B_3 = e^{1/2}N^2(2k+1)\sigma_r^2 \max(\gamma, B_{\lambda^{-1}}B_\lambda)^2$.

Our strategy is to choose all the Gaussian parameters from the last three conditions and select q accordingly using the first condition. However, since all the Gaussian parameters σ_r , σ_s , and σ_f are convolved in the conditions for adaptive hint problems above, it is hard to formulate the optimization problem to minimize these Gaussian parameters. Therefore, we first heuristically obtain the Gaussian parameters by ignoring insignificant terms and then check if they satisfy the security conditions afterwards.

We first investigate the condition for the last adaptive Hint-MLWE problem (4). Observe that $\sigma_1^2 \leq \sigma_f^2/\gamma^2$ by definition, we obtain

$$\begin{aligned} \sigma^{-2} &\geq 2(\gamma^2 + 2B_3Q)\sigma_f^{-2} \approx 4B_3Q\sigma_f^{-2} \\ &= (4e^{1/2}N^2(2k+1)\max(\gamma, B_{\lambda^{-1}}B_\lambda)^2Q)\sigma_r^2\sigma_f^{-2}. \end{aligned}$$

Based on this inequality, we set σ_f the smallest possible value as

$$\sigma_f = \sqrt{4e^{1/2}N^2(2k+1)\max(\gamma, B_{\lambda^{-1}}B_\lambda)^2Q} \cdot \sigma_r.$$

Now, by substituting this into the conditions for the first two adaptive-Hint MLWE problems (2 and 3), we obtain the following equalities:

$$\begin{aligned} - \sigma^{-2} &= 2(\sigma_s^{-2} + 0.5B_{\lambda^{-1}}^2 \max(\gamma, B_{\lambda^{-1}}B_\lambda)^{-2}\sigma^{-2}) \\ - \sigma^{-2} &= 2(1 + 2B_2n / (4e^{1/2}N^2(2k+1)\max(\gamma, B_{\lambda^{-1}}B_\lambda)^2Q\sigma^2)) \cdot \sigma_r^{-2} \end{aligned}$$

Since the first equality does not contain any unknown variables, we obtain an explicit formula for $\sigma_s := \sqrt{2 / (1 - B_{\lambda-1}^2 / \max(\gamma, B_{\lambda-1} B_\lambda)^2)}$. Then, as B_2 can be computed with known constants and σ_s , we also get the value for σ_r by substituting σ_s into the second equality. Recall that we used an approximate equality when computing σ_f , which may have reduced the size of σ_f . Therefore, we check if the condition for the final adaptive Hint-MLWE problem (4) actually holds, *i.e.*, the resulting σ is larger than 3.2. If not, we can simply increase the size of σ_f by setting it $\sqrt{2(\gamma^2 + 2B_3 Q)} \cdot \sigma$ without changing σ_r and σ_s , since a larger σ_f results in an increased security level for the first two adaptive Hint-MLWE problems (2 and 3), with larger σ . We note this last condition will likely hold nevertheless, as we already use a naïve bound σ_f/γ for σ_1 . Finally, it remains to choose the smallest q that satisfies the first condition.